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From participatory process to robust decision-making: An Agriculture-water-energy nexus analysis for the Souss-Massa basin in Morocco

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ABSTRACT

The Water-Energy-Food (WEF) framework is widely used to address sustainability and resource management questions. However, many WEF methods miss engaging with stakeholders in the process. In this study, we introduce a stakeholder-driven and model-supported robust nexus decision-making framework. This methodology is exemplified by a case study in the Souss-Massa basin (SMB) which has significant importance for the agricultural sector in Morocco. However, the water scarcity exacerbated by climate change, overexploitation of groundwater and heavy use of fossil fuels for pumping is threatening the future of this fertile land. An integrated agriculture, water and energy model was developed to explore various potential solutions or scenarios such as desalination, wastewater reuse and improved water productivity. The analysis revealed that engaging with stakeholders and developing common robust nexus decision metrics is essential to establishing a shared and transparent approach to address the complicated nexus challenges. It also showed that no one solution can address all nexus challenges and highlighted the need for an integrated strategy that stimulates the contributions from different sectors. Finally, the transition from fossil fuel groundwater pumping to solar pumping is shown to be economically and environmentally viable.

Introduction

The Water-Energy-Food (WEF) nexus framework was developed to guide the development of cross-sectoral policies, maximizing synergies, and minimizing trade-offs between water, energy, and food systems (Albrecht et al., Apr. 2018). The WEF nexus can be seen as a substitute for the traditional resource governance that was mostly focused on single resource categories (Bleischwitz et al., Dec. 2018). Such an integrated approach is not important only for resource management, but also for achieving the Sustainable Development Goals (SDGs) (Castor et al., Oct. 2020) (Benites-Lazaro et al., Nov. 2018). Managing and governing interlinkages between sectors and resources is key to achieving the SDGs such as SDG2 on zero hunger, SDG6 on water and sanitation, SDG7 on sustainable energy and SDG 12 on sustainable resource management, to name a few (Bleischwitz et al., Dec. 2018; Sindico, 2016).

In the last two decades, the WEF nexus concept has gained significant attention both in the academic and policy-making arenas. Several methodologies were developed to facilitate the implementation of this concept. A comprehensive compilation of methods for nexus assessments was performed by (Albrecht et al., Apr. 2018; Ramos et al., Dec. 2020), and a summary of the analytical tools used in nexus studies was given by the UNECE (United Nations Economic Commission for Europe. UNECE, 2018). To address the physical and social aspects of

Abbreviations: CLEWs, Climate, Land-use, Energy and Water strategies; DCS, Decarbonization strategies scenarios; DEM, Digital Elevation Model; DES, Desalination Scenario; FAO, Food and Agriculture Organization of the United Nations; GHG, GreenHouse gas; GIS, Geographic Information Systems; GWh, Gigawatt-hour; IEA, International Energy Agency; INS, Integrated Strategies scenario; IWP, Increased Water productivity scenario; IWRM, Integrated Water Resource Management; kW, Kilowatt; kWh, Kilowatt-hour; MABIA, MAitrise des Besoins d'Irrigation en Agriculture; MAD, Moroccan Dirham; MCM, Million Cubic Meters; NGOs, Non-Profit Organizations; ORMVA, The Regional Office for Agricultural Development in Souss-Massa; PM, Participatory modelling; PV, Solar Photovoltaics; RDS, Robust Decision Support; REF, Reference Scenario; SC, sub-catchments; SDGs, Sustainable Development Goals; SEI, Stockholm Environmental Institute; SMB, Souss-Massa basin; TBNA, Transboundary Basin Nexus Assessment; UNECE, United Nations Economic Commission of Europe; WEAP, Water Evaluation And Planning system; WEF, Water-Energy-Food; WWR, Wastewater reuse scenario; XLRM, eXegenous, Levers, Relationships and Metrics of performance.

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water, energy, and food systems, mixed-method approaches that combine quantitative and qualitative methods from multiple disciplines are needed (Albrecht et al., Apr. 2018). Common examples of nexus approaches that combine quantitative and qualitative methods include the Climate, Land-use, Energy and Water strategies (CLEWs) framework (Howells et al., 2013), the Food and Agriculture Organization of the United Nations (FAO)-WEF approach (Flammini et al., 2014), the WEF Nexus Tool (Daher & Mohtar, Sep. 2015), and the Transboundary Basin Nexus Assessment (TBNA) methodology developed by the United Nations Economic Commission of Europe (UNECE) (United Nations Economic Commission for Europe. UNECE, 2015; de Strasser et al., Feb. 2016).

Nexus approaches have different preferences when it comes to the implementation of mixed methods. For example, the focus of the CLEWs frameworks is to quantitatively analyse the interactions between the systems of climate, land, energy and water using different modelling tools (Ramos et al., Dec. 2020), while the TBNA methodology is a pragmatic and participatory driven approach that looks at stakeholders' participation not only as a useful tool to better analyse the nexus but as a necessary step to officially validate results and ensure the policy relevance of the assessment (de Strasser et al., Feb. 2016). Similarly, the FAO-WEF nexus approach relies on stakeholder engagement to develop a shared understanding of the context in which future interventions will be embedded and to ensure that these interventions are aligned with the national needs and priorities (Flammini et al., 2014).

However, frameworks with a balanced focus on both stakeholder engagement and analytical modelling tools, with clear processes to ensure robust decision making, are still needed. Albrecht et al. (Apr. 2018) argued a need to develop robust nexus methods that explicitly address the socio-political context, combine quantitative with qualitative data to advance interdisciplinary and mixed methods, and deeply engage with stakeholders and decision-makers. Therefore, this study aims at developing a stakeholder-driven and model-supported robust nexus decision-making framework by integrating different quantitative and qualitative methods and tools. This novel approach will then be exemplified in a case study of the Souss-Massa basin (SMB) in Morocco.

Case-study: the Souss-Massa Basin

The Souss-Massa province is situated in the middle-western part of Morocco on the Atlantic ocean. It has a total surface area of 27,000 km^2 and is home to 2.56 million people. The Souss-Massa region is among the top Moroccan agricultural zones (Ministry of Interior-Morocco. RMMI, 2015). It is estimated that almost 175,500 ha of the area in Souss-Massa is dedicated to the production of crops, including cereals, citrus, almond trees, and vegetables (MAPM, 2017). Souss-Massa produces 85 % of the vegetables and more than half of the exported citrus. Therefore about 50 % of the region's workforce is employed by the agriculture sector (Choukr-Allah et al., 2017) which, along with tourism and fishing, produces 6.6 % of Morocco's total GDP (Kingdom of Morocco High Commission for Planning, 2014).

Water resources are central for almost every socio-economic activity in Souss-Massa. Irrigation represents around 94 % of total water demand and the remainder goes for industry, tourism and municipal uses (Agence Francaise de Developpement, 2012). Groundwater aquifers supply about 70 % of the total demand estimated at 650 million cubic meters (MCM) plus 260 MCM above the recharge requirements on an annual average (Hssaisoune et al., 2016). Overexploitation of the groundwater resource and low rainfall levels, around 250 mm per year in the plain area, is constantly reducing the water table levels of the aquifers (Choukr-Allah et al., 2016). For example, the main aquifer in the basin, the Souss aquifer, has witnessed a drop in the water table from 15 m to 30 m, or 0.5–2.5 m per year in the last four decades (Brahim, 2016). The decline in the water table results in seawater intrusion, increasing pumping costs and compromising the water supply security (Moha et al., 2017).

In addition to groundwater resources, the SMB has 8 large and 16 small dams (Fig. 1) to control and store the surface water that is used for irrigation, drinking, industrial and protection purposes with a combined capacity of about 800 MCM (Choukr-Allah et al., 2017).

Morocco is a net energy import country; it relies on coal, oil and gas imports for most of its energy needs. This is also reflected in its electricity generation mix which is heavily dependent on coal (54 %), natural gas (19 %) and oil (9 %). The National Energy Strategy of Morocco aims at reaching 52 % of renewable electricity by 2030. Under the Paris Agreement, Morocco is committed to reducing greenhouse gas (GHG) emissions by 17 % below business-as-usual (BAU) levels by 2030, with an additional 25 % reduction by 2030 if international support is available, taking the total to 42 % below BAU (International Energy Agency. IEA, 2019). Active power plants in Morocco are concentrated in the northern part of the country. The SMB is an energy-dependent region and relies on the national grid and electricity generated in the power plants outside the basin to meet its demand. As mentioned previously, there are numerous dams in the area; however, they are used for irrigation and drinking purposes and none of them is used for electricity generation (Morocco Industrial Map, n.d.), even though hydropower comprised 4 % of the total electricity generation mix in Morocco in 2017 (International Energy Agency. IEA, 2019).

The use of heavily subsidized butane for groundwater pumping in Souss Massa is adding an enormous burden on the government's budget. According to the Directorate of energy and mines in Souss Massa, in 2019 the total consumption of butane for groundwater pumping reached 84,000 t. Recent estimates show that about 20 % of the pumping activity is powered by butane-driven pumps, while Solar Photovoltaics (PV) make up to 10 % and the remaining 70 % of pumps are powered by electricity from the national grid (consultation with local energy experts and The Regional Office for Agricultural Development ("ORMVA"[1]) in Souss-Massa.

Previous studies on Souss-Massa such as (Choukr-Allah et al., 2017; Brahim, 2016; Hirich et al., 2015)–(Malki, 2016; Mensour et al., 2019; Azim et al., 2017) were sector-centred. No study tackled the challenges in Souss-Massa from the WEF nexus perspective to date. This study contributes to filling this literature gap by implementing, for the first time in Souss-Massa, the WEF nexus approach to address sustainability challenges.

Methodology

A mixed methods approach is followed in this study integrating qualitative and quantitative methods. Among the several definitions of the mixed method research that evolved in the last three decades (Johnson et al., Apr. 2007) (Creswell, 2010), the definition used in this study is based on Johnson et al. (Apr. 2007) which defines the Mixed methods research as 'the type of research in which a researcher or team of researchers combines elements of qualitative and quantitative research approaches'. Using mixed methods overcomes weaknesses of one method with strengths of the other (Akimowicz et al., Jul. 2018) (Schlör et al., Mar. 2021). The mixed methods approach assumes that both analytical techniques complement rather than oppose each other. Therefore, it offers valuable new insights in complex issues by giving different perspectives of the question in hand (Schlör et al., Mar. 2021). Fig. 2 gives an overview of the mixed methods used in this analysis. The qualitative analysis drew upon the UNECE nexus method (United Nations Economic Commission for Europe. UNECE, 2018) (de Strasser et al., Feb. 2016) and the Robust Decision Support (RDS) (Purkey et al., Jul. 2018) (Huber-Lee et al., Mar. 2020). Both

[1] ORMVA: Office Régional de Mise en Valeur Agricole du Souss Massa.

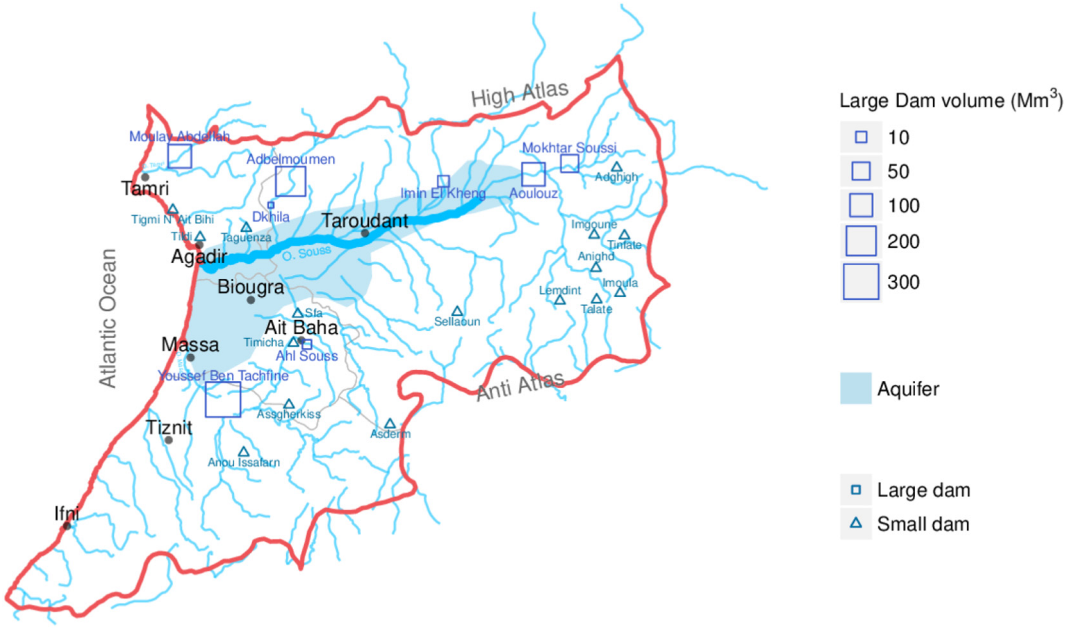

Fig. 1. Water resources of Souss-Massa River Basin.

have an extensive stakeholder engagement process that ensures that the quantitative modelling responds to the needs of the participants and policymakers. The quantitate analysis consists of two modelling tools: a Geographic Information System (GIS) based energy model and an agriculture and water model.

Qualitative analysis

Stakeholder engagement

The experiences of end-users bring valuable insights to the nexus studies (Yoon & Saurí, Dec. 2019). Engaging stakeholders early in the process ensures the complexity of the nexus – both the problems and the potential solutions – is captured in a policy-relevant context (Prell et al., Sep. 2007). This is particularly true given the deep uncertainties around climate change (Sharmina, 2019), but also other uncertainties, such as the ongoing pandemic that has profoundly affected a myriad of cascading factors across socio-ecological systems.

Furthermore, the multiple scales and conflicting understanding of stakeholders involved in natural resource management calls for a participatory process in integrated modelling (Prell et al., Sep. 2007).

Participatory modelling (PM) is an engaged research methodology where researchers work closely with stakeholders and center their knowledge and expertise in modelling complex systems (Quimby & Beresford, Feb. 2022). Unlike traditional problem-solving which is based on 'factual knowledge' and optimization modelling approaches, PM can be seen as stakeholder-based policy design and implementation (Prell et al., Sep. 2007). Especially in situations where consensus is low, uncertainty is high, and yet collaboration, co-decision, and joint action are needed, PM is well equipped to navigate through such complex situations (Kimmich et al., Dec. 2019) (Anggraeni et al., Nov. 2019).

As Kimmich et al. (Dec. 2019) argue, PM is relevant for the WEF nexus studies for three reasons. First, there is limited consensus on the methods applied, conflicting interest and scientific uncertainty. Second, there are evolving computational modelling frameworks to operationalize the WEF nexus in decision making. Third, the WEF nexus governance often calls for cooperation and adaptation between a large set of stakeholders coming with their different interests, uncertainties, politics and power dynamics (Kimmich et al., Dec. 2019).

In this study, a novel approach for interaction with stakeholders was designed by integrating two frameworks; the participatory process of

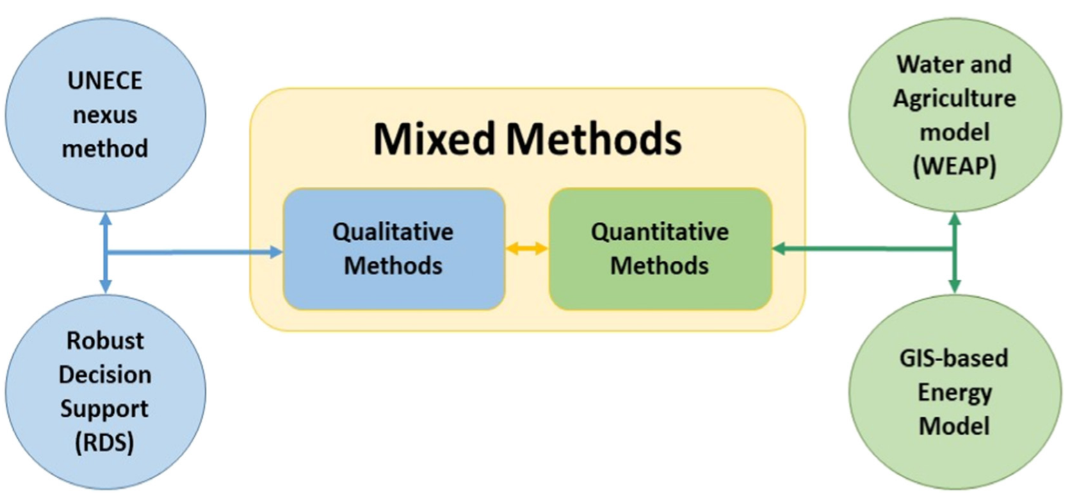

Fig. 2. Overview of the mixed methods implemented in this study.

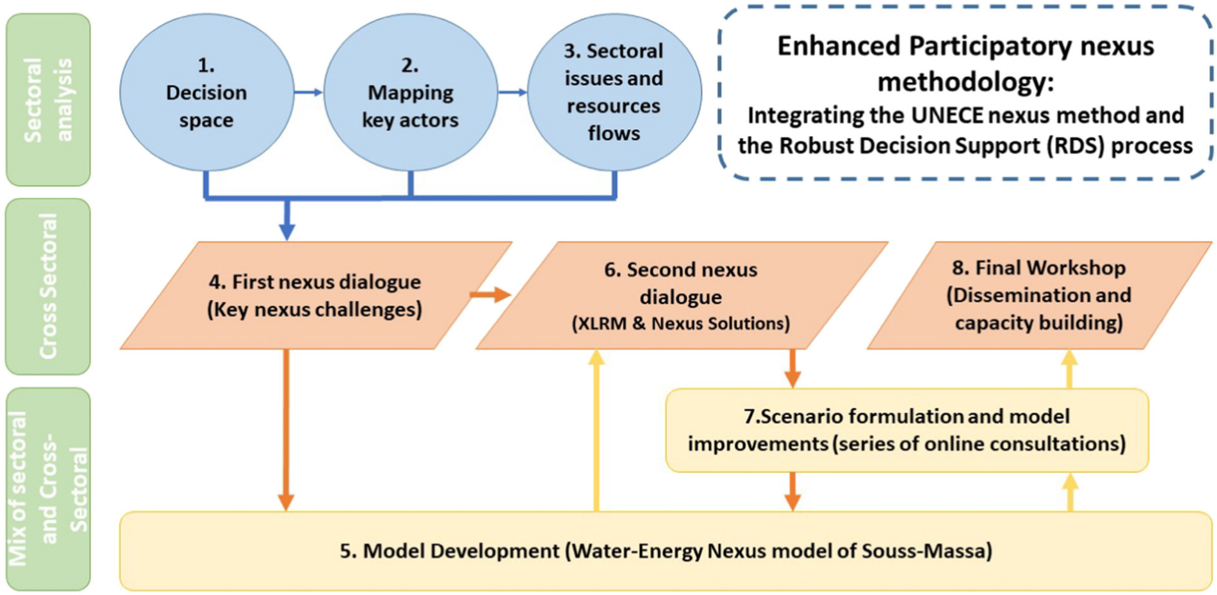

Fig. 3. Schematic diagram of the enhanced participatory nexus methodology.

UNECE nexus methodology and the Robust Decision Support (RDS) process from Stockholm Environmental Institute (SEI) and RAND corporation (Lempert et al., 2003) as illustrated in Fig. 3. The RDS started the exploration of sector-specific goals, critical uncertainties and policy/infrastructure strategies, while the UNECE followed by grouping stakeholders across sectors and guiding them to see how the sectors had intended and unintended implications across sectors.

Some changes were made to both frameworks to fit the purpose of the study and the subnational scale of application in the SMB. Unlike other studies that were on a transboundary level and involved more than one country (UNECE, 2017, 2020) this study can be considered as the first study to implement the UNECE methodology on a subnational level and to integrate it with the RDS process.

The methodology implemented in this study encompasses eight steps. **In the first step**, the decision space was identified and the SMB socioeconomic context was studied including the current state of the sectors of interest, main strategic goals and development policies. **The second step** focused on mapping the key actors (in each sector) involved in the process. **The third step** zoomed into the sectoral issues and resource flows using quantitative indicators (wherever possible) to help clarify their relative importance. Steps 1 to 3 focused on the sectoral perspective and were conducted in a "desk study" format, which serves as preparatory material for the next step. **The fourth step** was the **first nexus dialogue** with stakeholders or the 'problem formulation workshop'. Representatives from the agriculture, water, energy and environment sectors coming from different backgrounds (i.e. government, academia, Non-Governmental Organizations (NGOs) and private sector) interacted during a two-day workshop. The main outcome of this step was having a common understanding of the WEF nexus approach and mapping the 'key nexus challenges'[2] in the SMB. See more detailed description of this step in Appendix A. **The fifth step** used analytical modelling tools to study more in-depth the nexus interdependencies, and explored various solutions. A detailed description of the modelling tools and setup is given in Quantitative analysis: the modelling framework. **The sixth step**, or the second nexus dialogue, was conducted a year after the first one and took the stakeholders through the e**X**egenous, **L**evers, **R**elationships and Metrics of performance (**XLRM**) **process** (Lempert et al., 2003) to formulate potential solutions and to design the metrics used to evaluate the robustness of the solutions. **The seventh step** was formulated as a series of 'online consultation meetings and seminars' and running ensembles of scenarios to meet the metrics of performance developed in the previous step. **The eighth step** was the final dissemination of the findings coupled with a capacity building activity on the analytical tools.

The nexus dialogue explained in step four is a pivotal step in the methodology since it defines the key challenges. It starts with a presentation of the latest sectoral policies and plans, followed by an introduction to nexus methodology and how to identify the 'nexus challenges'. Participants were then split into groups to map the interlinkages between sectors and identify the 'nexus challenges'. Several challenges have been identified which make it critical to agree upon the priorities among stakeholders. After the first day discussions, a voting technique was introduced using an interactive tool called 'mentimeter'[3] that allows participants, among other things, to prioritize the identified challenges. This voting step is an addition to the traditional UNECE methodology and the Robust Decision Support (RDS) process, and its added value is co-creating a common understanding among stakeholders and giving the modelling team clear priorities for exploring the solutions.

XLRM process and scenario description

The four dimensions of the XLRM process were implemented as follows:

- X (e**X**egenous factors): represents the factors outside the control of the actors but which have the potential to influence the outcomes. The vulnerability assessment or the (vulnerability-Impact) mapping exercise was conducted with stakeholders to identify the vulnerability of the system to the key factors and their relative impact.
- L (**L**evers): represents the specific actions that are available to the actors as they seek to improve conditions or outcomes in the face of future uncertainty. Through stakeholders consultation, a number of scenarios were designed incorporating different levers as described below.
- R (**R**elationships): the models used to develop the relationships between the sectors. See Quantitative analysis: the modelling framework.
- M (Metrics of performance): by which actors will evaluate the outcomes of a specific scenario. Four key metrics were selected to

[2] By the 'key nexus challenges' we mean the challenges that have implications beyond the sector they are originating from.

[3] www.mentimeter.com an online tool to interact with audience during presentations and meetings.

evaluate the robustness of the scenarios: a) the impact on groundwater level, b) the impact on unmet water demand[4], c) crop production and d) energy demand.

The nexus dialogues and the XLRM exercise with stakeholders resulted in formulating six key scenarios to be explored using the water-energy model:

1. **Reference scenario** (**REF**): represents the current status in Souss-Massa. The total irrigated area is assumed to stay constant at 125,482 ha (see Table B4) while the domestic demands are assumed to increase over time with a growing population at 1.4 % per year.
2. **Desalination scenario** (**DES**): In the first phase (2021), the new Chtouka desalination plant (Konrad-Adenauer-Stiftung. KAS, 2020; Kettani & Bandelier, Nov. 2020) starts operation at a capacity of 275,000 m^3 per day, of which 150,000 m^3 per day are allocated for domestic use in Agadir and 125,000 m^3 per day for irrigating agricultural perimeters in Chtouka. In the second phase (2030) the desalination capacity is expanded to 450,000 m^3 per day with an equal share between both regions.
3. **Wastewater reuse scenario** (**WWR**): This scenario assumes a combination of seawater desalination and wastewater reuse to explore the impact of reusing the treated wastewater for irrigation purposes. In addition, part of the treated water from the Agadir wastewater treatment is reused within Agadir for landscape irrigation.
4. **Increased water productivity scenario** (**IWP**): This scenario explores mitigating irrigation demand through increased water productivity. This means advances in irrigation, fertilizer application, soil tillage, and farming practices to avoid the overuse of water.
5. **Integrated strategies scenario** (**INS**): This scenario combines different strategies by increasing water supplies (through desalination and wastewater reuse) and increasing water productivity.
6. **Decarbonization strategies scenarios** (**DCS**): On top of the above, several scenarios were explored for the decarbonisation of the agricultural sector, more specifically, to phase out butane used for groundwater pumping and to increase the share of solar pumping. Two criteria were selected to develop the sub-scenarios and run the sensitivity analysis:
   A) Butane phase-out year:
   - **None**: which means the use of butane will continue to the end of the modelling period (2050) at the same share of 20 % of the irrigated area.
   - **Late phase-out** (**by 2040**): the use of butane will gradually decrease until it is completely phased out by 2040.
   - **Early phase-out** (**by 2030**): the use of butane will gradually decrease until it is completely phased out by 2030.

   B) PV adoption level (in the total electricity demand for groundwater pumping):
   - 10 %: the current share of PV will continue to the end of the modelling period (2050).
   - 20 %: doubling the current share of solar pumping by 2040.
   - 40 %: aims at reaching a 40 % share of PV by 2040.
   - 60 %: aims at reaching a 60 % share of PV by 2040.

Quantitative analysis: the modelling framework

The quantitative analysis is based on two analytical modelling tools. The first is a GIS-based energy model (henceforth referred to as the energy model) that estimates in a spatially distributed manner the electricity requirements for different processes and explores several decarbonisation strategies. The second is the **W**ater **E**valuation **A**nd **P**lanning system (WEAP) which is a water system model of the SMB (henceforth referred to as the water model). The two models were soft linked to assess a number of strategies or scenarios for sustainable agriculture in the SMB.

Energy model overview

a) Model structure:

A GIS-based approach was chosen to develop the energy model to capture spatial differences such as elevation variation, groundwater depth and renewable energy sources in each location. Avoiding the aggregation of the special dimension provides more realistic and relevant insights for decision-makers. The full code is available on a GitHub repository (Gomez & Almulla, 2022) and a detailed description of the mathematical relations used in the energy calculations can be found in Appendix B. The methods and code were implemented by the authors on a similar case study in Jordan (Ramirez et al., Apr. 2022).

Different streams of inputs feed into the energy model as shown in Fig. 4. The inputs from the participatory process make the first group. This includes, for example, sectoral plans and limitations, the nexus challenges, key assumptions (especially when it comes to local data) and feedback on the scenarios. The second group is input from the water model, which includes: the schematic structure of the water system and the scenario run results. Then come a number of GIS layers such as administrative boundaries, cropland area, elevation, solar irradiations and others (see Table 1). The last group of inputs is related to the technical specifications of different technologies (such as costs, efficiencies, lifetime and others). These inputs are blended using the open-source programming language 'python' (Python Programming Language, 2001).

b) Estimation of the electricity requirements for different uses:

The first part of the energy model estimates the electricity requirements for four types of processes: groundwater pumping, surface water conveyance, seawater desalination and wastewater treatment. The model estimated the electricity demand (in kWh) for each location, for the entire modelling period 2020–2050 and for each scenario.

The groundwater pumping depends mainly on the volume of the water pumped in each month and the depth of the groundwater table. The first is estimated using the water model and the second is based on remotely-sensed data on groundwater levels obtained from online sources as shown in Table 1.

Similarly, the surface water conveyance depends, among others, on the volume of water being conveyed, the elevation differences and the pipeline type and diameter (Ahlfeld & Laverty, 2011). The water model estimates the monthly water conveyed while the digital elevation model (DEM) layer (de Ferranti, 2014) gives the elevation differences. The values of the pipeline diameters are based on literature and consultation with local stakeholders as shown in Table 2. Two major surface water conveyance pipelines exist. The first pipeline (line A in Fig. 5) conveyes water from 'Mohtar Soussi dam' to 'ElGuerdane' irrigation scheme. This line depends on elevation difference and no additional pumping is required. The second pipeline (line B in Fig. 5) conveyes water from 'Youssef Ben Tashefine dam' to the 'Modeen Public Massa' irrigation scheme. This is the only major pipeline that applies pumping. Electricity requirements for the distribution network are not considered in the analysis since they require relatively smaller amount of electricity and there is no clear map of the distribution network. The electricity requirement for desalination is based on the specifications of the Chtoka desalination plant (Konrad-Adenauer-Stiftung. KAS, 2020; Kettani & Bandelier, Nov. 2020) using a conservative average energy intensity of 3.31 (kWh/m3). For wastewater treatment, the water model includes 12 existing and planned wastewater treatment plants. To account for improving the quality of the treated wastewater for reuse purposes, a process with primary and secondary treatment was assumed, increasing the energy intensity of the facilities from 0.1 kWh/m^3 to 0.8 kWh/m^3

[4] Serves as a water scarcity indicator by measuring the gap (in percentage) between the water demanded in each sector versus the water that is actually delivered.

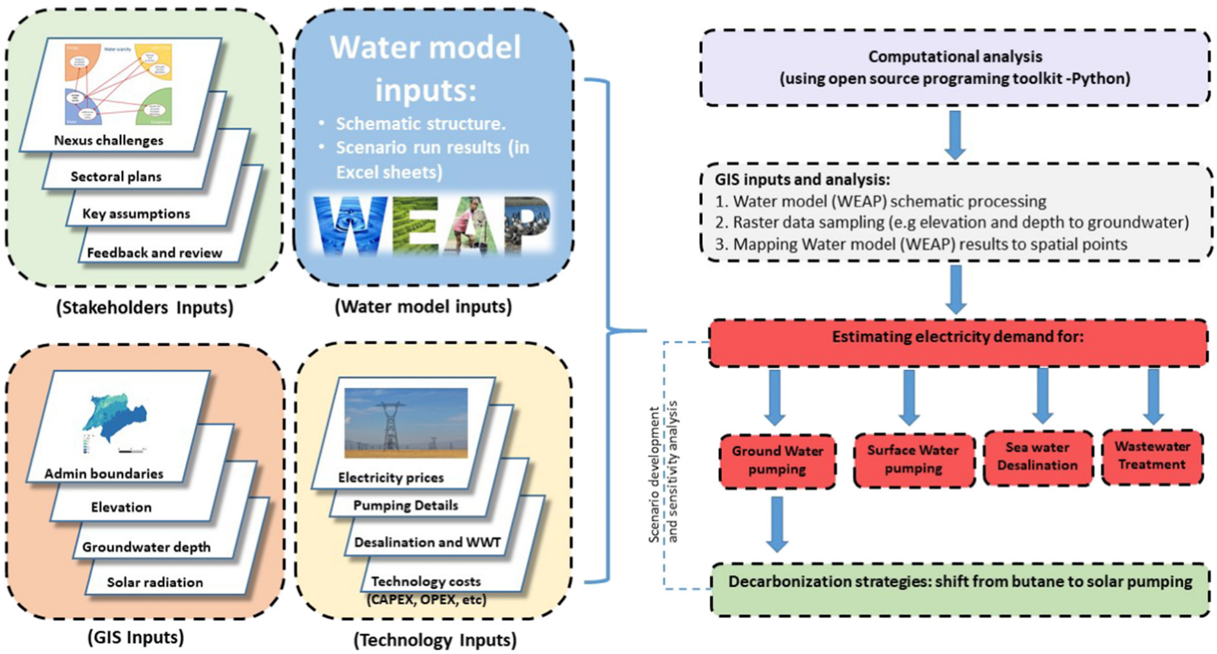

Fig. 4. Overview of the modelling structure showing the key inputs to the energy model and different energy modules.

(Barroso Soares, 2017). It is assumed that the discharge of the facilities is used for irrigating the agricultural perimeters near the location of the facilities, which results in negligible energy requirements for conveyance.

The energy and water systems are indispensable and the nexus between those two systems has another dimension which is 'water-for-energy' (Fayiah et al., Feb. 2020). For example, water can be used for cooling thermal power plants, to generate electricity in hydropower plants, or to clean solar panels. However, as mentioned earlier, currently no power plant exists in the SMB, therefore this study has not considered the water-for-energy dimension. It is nevertheless noteworthy that Morocco has plans to install a 350 MW and 1.3 million cubic meter pumped storage facility in the 'Abdelmoumen dam' located 70 km Northeast of Agadir city in the SMB (Ministry of Energy Transition and Sustainable Development, n.d.). The project will supply the peak load and will serve as a rapid-response solution to offset any drops from wind farms in southern Morocco (Vinci-construction, n.d.). The project will be using the water of the already existing dam developed for irrigation and drinking purposes. This would create competition or conflicts between irrigation, drinking and electricity demands that would be important to explore in future studies.

c) Decarbonization strategies for the agricultural sector

The second part of the energy model was developed in response to the request from the Regional Directorate of Energy and Mines in Souss-Massa (Direction Régionale de l'Energie et des Mines). The request was to explore different decarbonisation strategies related to phasing-out butane use in groundwater pumping. The model was set up to compare different alternatives to move away from butane and to increase the share of solar pumping. Currently, butane powers about 20 % of the pumps, while solar pumps cover only 10 %, and the remaining 70 % are powered by the grid. The comparison is based on economic and environmental metrics that are the total cost bearded by the government of Morocco (in million Moroccan Dirham 'MAD') and the CO_2 emission levels (in MtCO2). Table 3 summarises a number of inputs used in this part of the model.

Water model overview

The Souss-Massa WEAP model was developed to represent the main water supply and demand features of the basin. The spatial scale used in the model is appropriate to simulate major hydrologic flows, to represent major demographic trends and to evaluate the effects of water management responses to changes in the basin.

Table 1
GIS inputs used in the energy model.

#	Parameter	Format	Unit	Source
1	Souss-Massa borders	Vector (shp)	Polygon	Local sources (ORMVA).
2	Provincial borders	Vector (shp)	Polygons	Local sources (ORMVA).
3	Digital Elevation Model (DEM)	Raster (tif)	Meters above sea level	(de Ferranti, 2014)
4	Groundwater depth	Raster (tif)	Meters below ground level	(Fan et al., 2013)
5	Irrigated area extent	Vector (shp)	Polygons	Local sources (ORMVA).
6	Monthly solar irradiation	Raster (tif)	kJ/m^2day[1]	(WorldClim, 2016)

Table 2
Non-GIS inputs used in the energy model.

#	Parameters	Units	Value	Source
1	Transmission pipeline diameter	m	1.7	(Ratnayaka et al., 2009; Hassi et al., May 2021)
2	Desalination pipe diameter	m	1.5	(AFESD, 2014)
3	Desalination energy intensity (Reverse Osemosis)	kWh/m^3	3.31	3.5 as per (Konrad-Adenauer-Stiftung. KAS, 2020)
4	Wastewater treatment energy intensity (primary treatment - no reuse)	kWh/m^3	0.1	(Barroso Soares, 2017)
5	Wastewater treatment energy intensity (secondary treatment - with reuse)	kWh/m^3	0.8	(Barroso Soares, 2017)

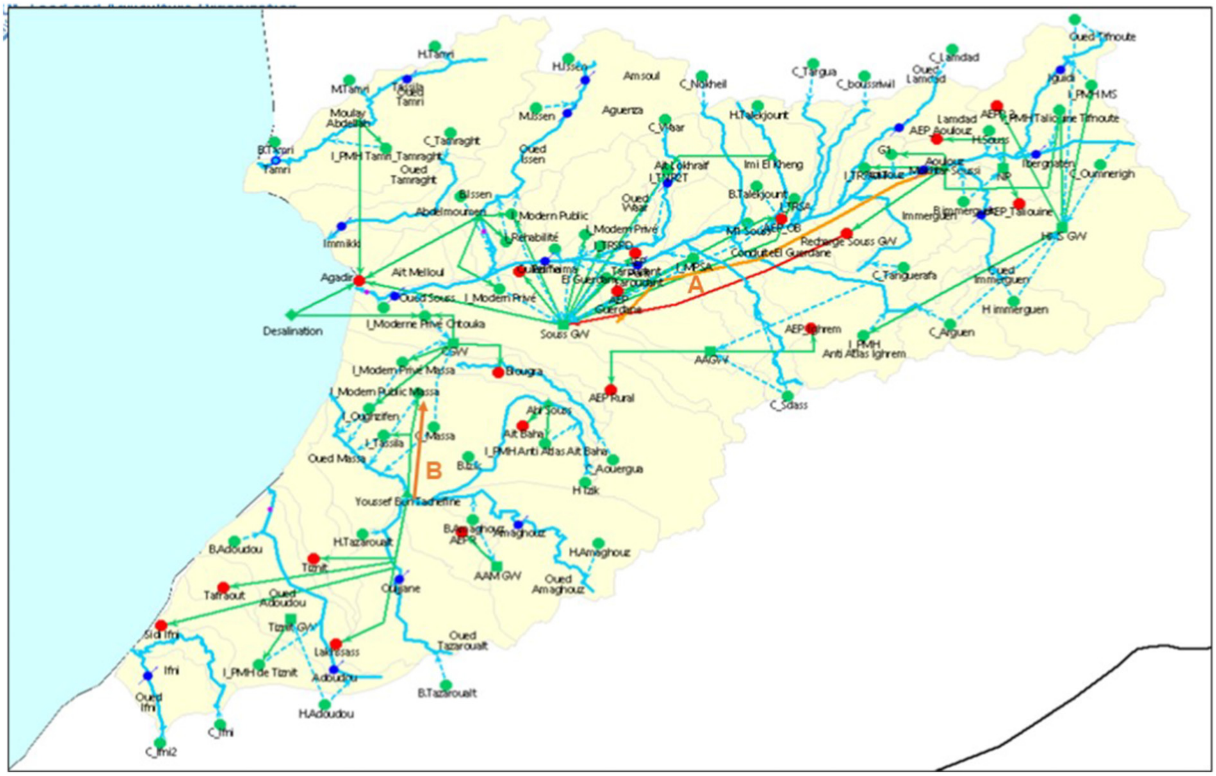

Fig. 5. WEAP Schematic of the Souss-Massa Basin Model.

The WEAP model is structured as shown schematically in Fig. 5 and consists of the following components:

- **Fifty sub-catchments** (**SC**) that simulate the basin hydrology (shown as green dots in Fig. 5). The built-in WEAP's rainfall-runoff soil moisture method is used to model the SCs and their contributions to streamflow and groundwater which makes up the watershed's hydrology.
- **Water demands**: which consist of
  - o **Municipal demand**: in 19 demand sites representing municipal water use in the main towns within the basin (shown as red dots). Water usage within each of these demand sites is split between household, administrative and industrial water use. The demand points are aggregated by the provinces of Agadir, Taroudannt, Inzegane-Ait-Mellul, Chtoucka-Ait-Baha and Tiznit.
  - o **Irrigation demand**: 22 irrigation districts are represented in WEAP. Cropped areas were estimated based on data obtained from local sources (ORMVA, see Table B4). Cropped areas were set for the four general crop types (cereals, fodder, trees and vegetables). The vegetables were divided into four main crops grown in the Souss-Massa basin (tomatoes, peppers, melons, and potatoes) which were further divided based on whether they are grown in greenhouses or open fields. To better represent the agriculture sector in the model, the MABIA[5] method to determe crop demands was utilized. The MABIA-WEAP package (Jabloun & Sahli, 2012) allows for daily simulation of evapotranspiration, irrigation needs and other climatic and crop-specific variables.

- **Water Supply**: which consists of
  - o **Surface water supplies**: include flows in wadis and storage in the six reservoirs distributed in the basin. In addition to the mainstream of the Souss and Massa wadis. The model simulates flows on the major tributaries to these wadis.
  - o **Groundwater supply**: groundwater from shallow and unconfined aquifers that are physically linked to catchments were added to the model as 8 major groundwater nodes.
  - o **Treated Wastewater**: Most communities within the Souss-Mass basin include wastewater treatment plants; however, the treated water is not being currently used as a water supply. This might change in the future to provide water for irrigation and other activities. As such, 12 existing and planned wastewater treatment plants were modelled in WEAP with a total existing capacity of about 79,000 m3 per day and a provision for additional 37,000 m3 per day (as shown in Table 4).
  - o **The Agadir desalination project**: which is being constructed to supply water to Agadir and to irrigate agriculture in the Chtouka region was also considered.

- **Connecting Water supplies and demands**: any demand node in WEAP can be connected to multiple water sources using transmission links (green lines in Fig. 5). The model decides which of these sources to draw from based on assigned preferences such as cost, water quality or other considerations.

The historical period of simulation for the model runs was set from 1976 to 2017, which support model calibration based on local data provided by local institutions. All future scenarios simulate the basin from 2020 to 2050.

The climate conditions affect the evapotranspiration and crop water requirement which will result in changes in agricultural water demand and consequently energy demand. Two climate conditions were implemented in the water model for each scenario:

- **Historical trends**: which represent the historical trends of evapotranspiration projected into the future.

[5] MABIA in French stands for "*MAitrise des Besoins d'Irrigation en Agriculture*." which translates as "*Control of irrigation needs in agriculture*".

Table 3
Key inputs and assumptions used in the energy calculations and butane phase-out scenarios.

#	Parameter	Value	Unit	Source
1	Cost of electricity production from Coal	0.65	MAD/kWh	Consultation with local experts (Food and Agriculture Organization of the United Nations. FAO, 2020)
2	Cost of electricity production from CSP	1.4	MAD/kWh	Consultation with local experts (Food and Agriculture Organization of the United Nations. FAO, 2020)
3	Cost of electricity production from PV	0.44	MAD/kWh	Consultation with local experts (Food and Agriculture Organization of the United Nations. FAO, 2020)
4	Avg Cost of electricity production from the national grid	0.57	MAD/kWh	Computed based on technology shares and costs.
5	PV capital cost	7000	MAD/kW	Consultation with local experts (Food and Agriculture Organization of the United Nations. FAO, 2020)
6	PV O&M Cost	1	% of the capital cost	Assumption
7	Butane subsidy	80	MAD/bottle (12 kg) of Butane	Consultation with the Regional Directorate of Energy and Mines in Souss-Massa. (Food and Agriculture Organization of the United Nations. FAO, 2020)
8	Butane consumption in agriculture in Souss Massa (2019)	84,000	Tons/year	
9	Butane use in Agriculture compare to total use of butane	36	%	
10	Share if Agricultural sector emissions (2016/2017)	5.4	% of total emissions	(International Energy Agency. IEA, 2019)
11	emissions from Electricity and heat producers	23	MT of CO_2	(International Energy Agency. IEA, 2019)
12	Total Emissions in Morocco (2017)	58	Mt of CO_2	(International Energy Agency. IEA, 2019)
13	Emission factor of electricity from grid	0.7	$kgCO_2$/kWh	computed based on data from (International Energy Agency. IEA, 2020)
14	Butane emission factor	3.1	kgCO2/kWh	computed based on data from (EPA, 2018)
15	Electric pumps efficiency	45	%	Consultation with local experts (Food and Agriculture Organization of the United Nations. FAO, 2020)
16	Butane driven pumps efficiency	20	%	Consultation with local experts (Food and Agriculture Organization of the United Nations. FAO, 2020)

- **Extended droughts**: which capture historical drought cycles and projects them into the future to represent a dry climate trend.

The GIS-based schematic structure of the water model enables the soft-linking of the water and energy models. This was done by running a pre-processing code that: a) transfers WEAP schematic with the energy layers (as illustrated in Fig. 6), b) samples a number of raster datasets (such as elevation and groundwater depth) to the Souss-Massa boundaries, and c) maps the WEAP results to the spatial points of interest.

Results and discussion

Results of the qualitative analysis

In the first nexus dialogue, the authors guided the stakeholders through the nexus challenges mapping exercise. In mixed groups from different sectors, the stakeholders identified the key interlinkages of the nexus challenges on different sectors. For example, and as shown in Fig. 7 the impact of water scarcity and droughts extends beyond the water sector. Droughts reduce groundwater table levels and increase the irrigation demand which as a result would require more energy for pumping. This means an increase in production costs, negative impact on the environment and overexploitation of the stressed water resources.

Table 4
Existing and planned wastewater treatment plants in the Souss-Massa basin.

Community	Capacity (m^3/day)		Irrigation perimeter
	Existing	New	
Agadir	62,430		Moderne Public Massa
Ait Baha	398	520	PMH Anti Atlas Ait Baha
Ait Iazza	1100		Moderne Public Souss Amont
Aoulouz		675	Traditionel Rehabilite du Souss 1ere
Biougra	1156	3800	Moderne Prive Massa
Drargua	1000	3335	Traditionel Rehabilite Issen
El Guerdane		935	El Guerdane
Lqliaa		9800	Moderne Public Massa
Oulad Berhil	1530		Traditionel Rehabilite Souss Amont
Oulad Teima	6000	8000	Moderne Public Issen
Tafraout	312	661	N/A
Tiznit	4900	9104	N/A
Total	78,826	36,830	

Similarly, a number of nexus challenges have been mapped by stakeholders (see Appendix C). The second step was to reach an agreement on the priority challenges. An online voting tool 'mentimeter' was used and participants voted anonymously to prioritize the challenges as follows:

1. Water scarcity and droughts (scored: 4.4 out of 5),
2. Energy sufficiency and independence (3.8),
3. Agricultural productivity (3.6),
4. Water quality and the need for desalination (3.5).

The importance of this collaborative mapping and prioritizing of the nexus challenges goes beyond realizing the multi-sectoral impact of the challenge in question. Using a participatory approach builds consensus among stakeholders on the potential implications of the issues as well as on the solutions. It helps stakeholders to realize how their sectoral strategies might exacerbate or mitigate risks in other sectors. The nexus framework gives equal weight or importance to all sectors, unlike other frameworks e.g. the Integrated Water Resource Management (IWRM) that looks to other sectors as water users only (Almulla et al., 2018). As mentioned earlier, this was the first time such a participatory nexus framework was implemented in SMB and Morocco in general, which can be considered a valuable science-based contribution to inform sustainable resource management in the basin that would otherwise be difficult to achieve.

The (vulnerability-impact) mapping was used to identify the vulnerability of the WEF system in SMB to exogenous factors (X, in the XLRM approach) that are -in most cases- outside the control of decision-makers. After a group brainstorming and discussion, each team was asked to use the (Uncertainty-Impact matrix) to map and set the relative importance of each factor based on the Uncertainty level (high or

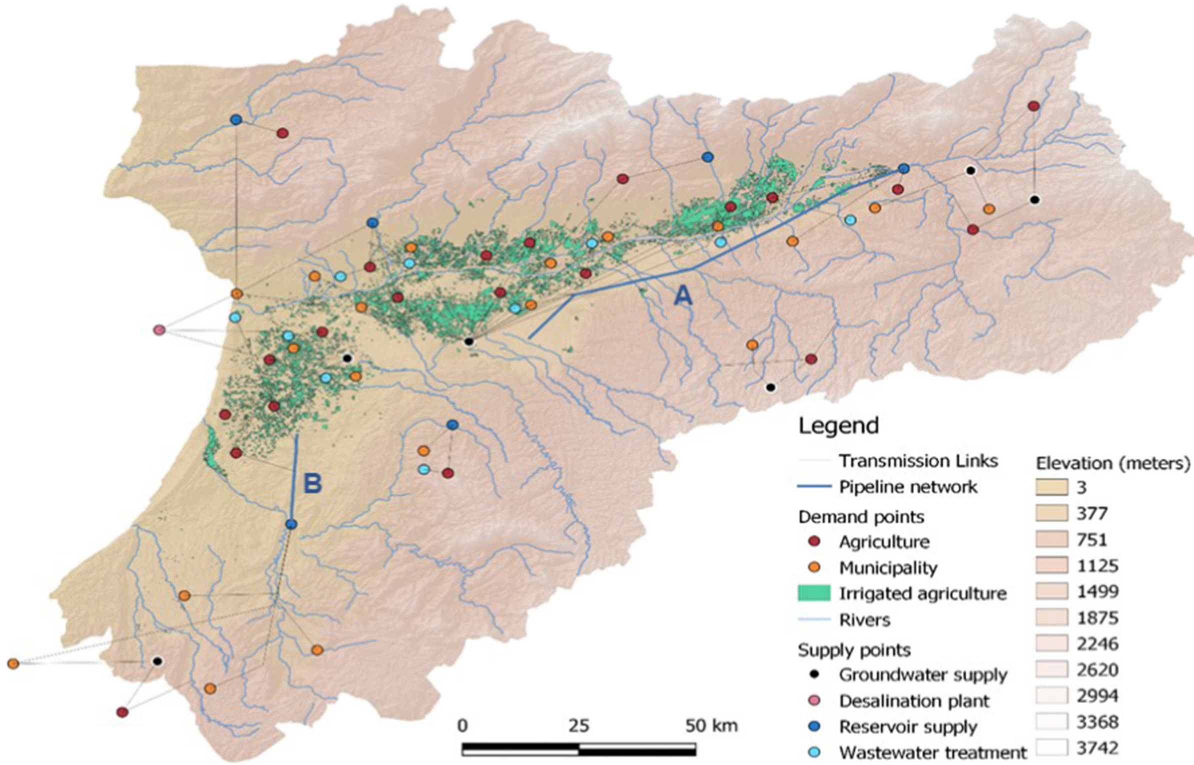

Fig. 6. Illustration of the water and energy models integration for the Souss-Massa basin.

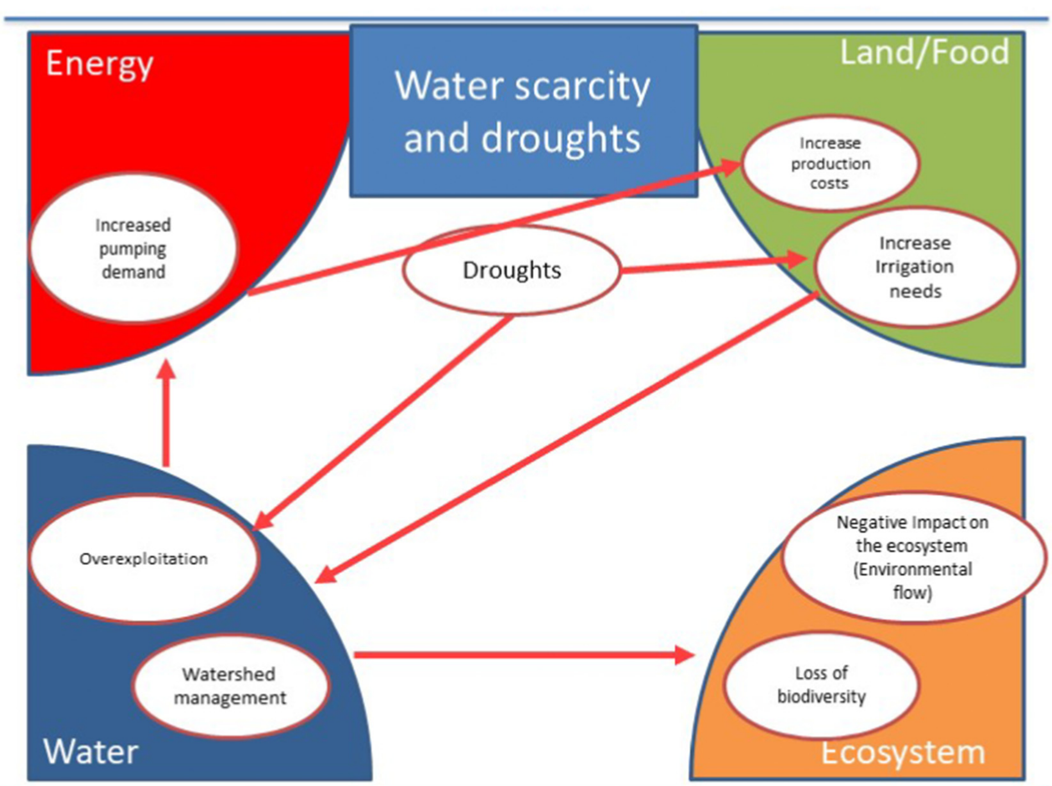

Fig. 7. Mapping of the nexus challenge - water scarcity.

low) and the expected Impact level (high or low). Results of the stakeholder discussions are shown in Fig. 8.

The factors that show a high level of uncertainties and impact were climate change, cost of energy imports, drop in water table level, change in international markets for agricultural products and change in consumers behaviours, while the factors that show relatively lower uncertainty but still high impact were the change in demography, change in agricultural practices, degradation of natural resources, technologies costs and changes in irrigated areas. Selected aspects (such as technologies costs, water table levels and cost of energy products) were carried forward into the quantitative analysis to test the sensitivity of the system to these parameters. The selection was based on what can be quantified using the modelling tools.

Integrating the robust decision-making approach using XLRM into the traditional UNECE nexus dialogues provided valuable insights that the single methodology could not. Stakeholders were able to jointly map the key uncertainties and evaluate their impact on the nexus system.

Results of the quantitative analysis

This section discusses the modelling results based on the four metrics as follows:

A. Impact on groundwater level:

Results show that if the current situation continues without any intervention, groundwater levels will continue declining. The level of the Souss aquifer (the main aquifer in the region) will drop from 170 m in 2020 to 180 m by 2050, while the Chtouka aquifer will drop from 84 m to 101 m in the same period. These values correspond to average decline in the water table level of 0.3 m per year in the Souss aquifer and 0.6 m per year in the Chtouka aquifer. Which are relatively lower than the historical annual decline of 1.6 and 1.1 m per year in Souss and Chtouka aquifers respectively between 1996 and 2010 (Choukr-Allah et al., 2017). The dry climate will result in a sharper decline which will have implications in terms of increasing pumping demand and energy bills for farmers.

The introduction of the desalination unit (DES scenario) will improve water availability for agricultural purposes, which will reduce the stress on groundwater aquifers, especially the Chtouka aquifer. As shown in Fig. 9, the water table level in Chtouka aquifer will stabilize at the level of 90 m instead of 101 m in the REF scenario and reach 94 m by 2050 at its lowest. The impact on other aquifers, including the Souss aquifer, will be marginal. This can be attributed to the fact that the desalinated water will be used for irrigation in Chtouka province only.

The reuse of the treated wastewater (WWR scenario) shows an interesting positive trend from 88.6 m depth in 2030 to 87.8 m in 2040 and reaching the lowest level of 90 m by 2050. The INS scenarios will further improve the water table level in Chtouka aquifer and will reach the level of 86.6 m by 2040 and 87.4 m by 2050 as shown in Fig. 9. This is due to the accumulation of benefits from all strategies assumed under this integrated scenario.

B. Impact on unmet water demand

Under the reference scenario, the unmet water demand for the agriculture and municipal sectors will increase in the coming decades, reaching the level of 60 % and 9 % respectively in the last decade (2040–2050).

The desalination project will provide a new source of water which will lower the level of unmet water demand for agricultural purposes from about 60 % to 45 % by 2050 as shown in Fig. 10. The enhanced

Impact	Uncertainty: Low			Uncertainty: High			
High	Risk of low involvement of the key or concerned actors	Change in Agricultural practices	Degradation of natural resources	Fluctuation in the sales (market) of the agricultural products	**Drop in the water table level**	**Cost of energy (imports)**	**Climate Change**
High	Increased electricity cost (for farmers)	Extension of irrigation area	Demography			Consumers behaviour towards the use of treated wastewater	the international price of agricultural products concerning the cost of production
High	Research and development (positive impact)	Growth in energy demand (population + economic sectors)	**Cost of technology**		Change in consumers behaviour and its impact on the market		
Low					Political decision (at international level)		
Low				Introduction of new technologies			

Fig. 8. Results of the uncertainties mapping. The x-axis is the key uncertainty scale ranging from low (to the left) to high (to the right). The y-axis is the impact scale ranging from low (at the bottom) to high (at the top).

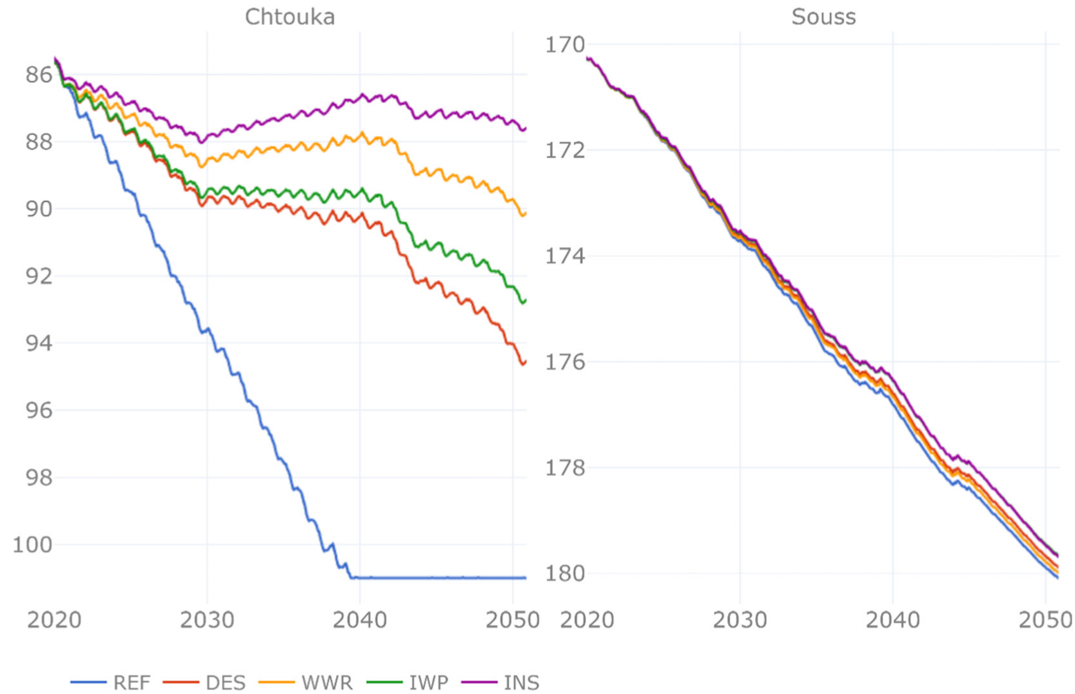

Fig. 9. Change in groundwater table depth (m) for Chtouka and Souss aquifers across all scenarios: Reference (REF), Desalination(DES), Desalination with Wastewater Reuse (WWR), Increased Water Productivity (IWP) and Integrated Strategies (INS).

farming practices assumed in the IWP and INS scenarios will improve water efficiency which will further lower the unmet water demand level to 34 % by 2050 or about 5 % average improvement in unmet water demand in the last decade. The gains in terms of domestic unmet water demand are very small since most of the domestic needs are being prioritized and supplied.

C. Impact on agricultural productivity

Water scarcity due to climate change would negatively affect agricultural production, especially in the last decade when the water shortages are the highest as shown under the reference (REF) scenario graph in Fig. 11. The drought conditions will exacerbate the situation and less water will be available for crops, hence less production. The change in crop production for different types of crops was modelled using WEAP.

The better availability of water resources due to the desalination project shown earlier will be reflected in crop production. The benefits in the last decade will reach about 43 % (compared to reference). The reuse of treated wastewater will have a minor impact but the IWP scenario, which assumes efficient water uses, will increase crop production by 57.5 % while the integrated scenario will further increase the production to almost 60 % in the last decade. From agricultural productivity (kg/m3) perspective, the productivity will increase from 1.6 (kg/m3) in the REF scenarios to 1.94 (DES, WWR) and 2.2 in (IWP and INS) scenarios respectively (see Table 5).

D. Impact on energy requirements

In terms of energy requirements, results show that the main electricity user is **desalination**. Energy is required for both water treatment processes within the desalination plant, as well as to convey water from the plant to the irrigation perimeters. The desalination plant will demand about 300 GWh of electricity every year, then increase to about 470 GWh per year in the second phase of the project in 2030. The second energy-intense activity is **groundwater** pumping which accounts for an average of 470 GWh per year as shown in Fig. 12. The

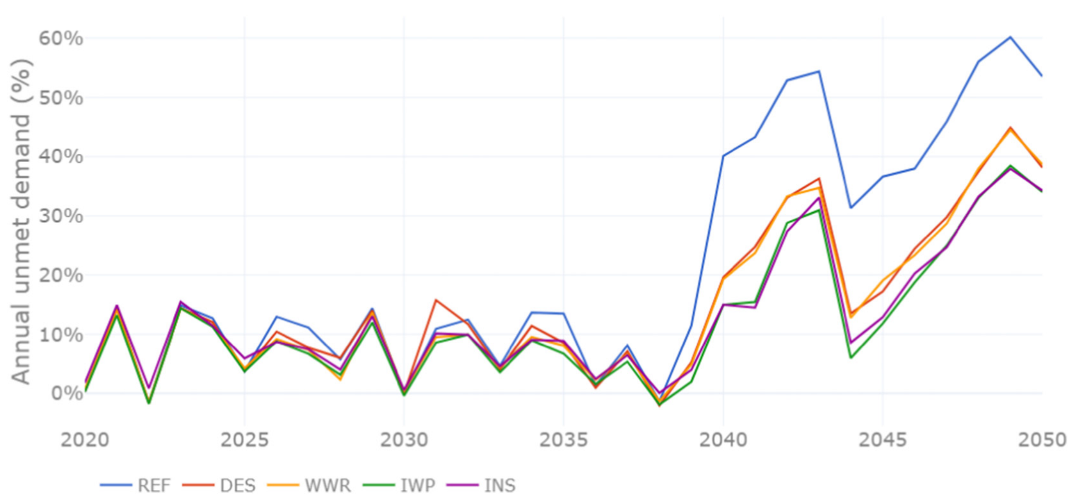

Fig. 10. Unmet water demand in agriculture (%) across all scenarios: Reference (REF), Desalination(DES), Desalination with Wastewater Reuse (WWR), Increased Water Productivity (IWP) and Integrated Strategies (INS).

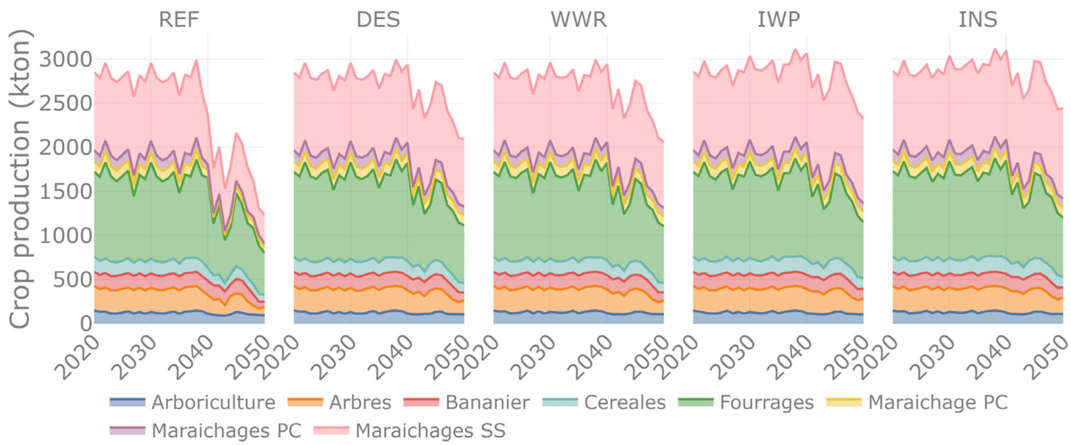

Fig. 11. Crop production (ton) across all scenarios: Reference (REF), Desalination(DES), Desalination with Wastewater Reuse (WWR), Increased Water Productivity (IWP) and Integrated Strategies (INS).

conveyance of desalinated water requires about 50 GWh in phase 1 and 140 GWh in phase 2. The surface water conveyance consumed relatively less energy with an annual average of 28 GWh reducing slightly under the WWR and INS scenarios. The **wastewater treatment** plants require also low energy since only primary treatment is assumed in the (REF, DES and IWP) scenarios. In the case of reusing the treated wastewater (WWR and INS scenarios), additional secondary treatment (activated sludge) is assumed (Food and Agriculture Organization of the United Nations. FAO, 1992) which will increase the energy demand for wastewater treatment from 4 GWh in REF to 35 GWh in the WWR and INS scenarios. The treated wastewater is assumed to be used in the agricultural area close to the treatment plants.

This explains the large increase in total electricity demand in the SMB in the desalination scenario (compared to the reference) and the almost similar profile in all other scenarios (WWR, IWP and INS), including desalination as well. It is worth mentioning that the annual fluctuation in electricity demand is due to changes in water requirements estimated by the water model.

The current plans show that the desalination plant will be powered by the Moroccan grid that is fossil fuel-based. According to the IEA, fossil fuels made up to 90 % of the Moroccan electricity generation mix in 2019 (International Energy Agency. IEA, 2020). The desalination plant would on the one hand benefit the water and agricultural sector, but on the other hand, would increase greenhouse gases since it is being powered by fossil based electricity. The Moroccan government has announced on several occasions that renewables (solar and wind) shall be used to power the desalination project (Guerraoui, Aug. 27, 2017; Herman, Jun. 18, 2019). However, there is no clear evidence on the ground yet, as confirmed by stakeholders.

The impacts of all scenarios are summarized in Table 5. No one solution can address all the nexus challenges. Therefore an integrated strategy that stimulates contributions of all actors, is needed to ensure the sustainable management of the resources. This would not be obvious without setting clear metrics to evaluate the robustness of nexus solutions. We argue that this is an important added value of integrating the RDS process and its XLRM tool with the UNECE participatory

Table 5
Summary results of the scenarios.

Scenario	Impact on groundwater level (m)*	Impact on unmet water demand (%)**	Impact on agricultural productivity (kg/m3)	Impact on energy requirement (GWh)***
Reference Scenario (REF)	Souss: -180m Chtouka: -101m	Agriculture: 46.6%	1.6	516
Desalination Scenario (DES)	Souss: -179.78m — Chtouka: -94m (-7.5m) ↓	29% ↓	1.94 ↑	1016 ↑↑
Wastewater reuse scenario (WWR)	Souss: -179.88m — Chtouka: -89.7m (-9.0m) ↓	28.7% ↓	1.942 ↑	1029 ↑↑↑
Increased Water Productivity scenario (IWP)	Souss: -179.55m — Chtouka: -92.3m (-11. 5m) ↓	23.4% ↓↓↓	2.19 ↑↑	1001 ↑↑
Integrated Strategies scenario (INS)	Souss: -179.56m — Chtouka: -87.3m (-13.7m) ↓↓↓	23.8% ↓↓	2.24 ↑↑↑	1007 ↑↑

**Last year (2050) value. **Last decade (2040-2050) avg. value ***Entire modelling period (2020-2050) avg. value*

(↑) Negative increase. (↑) Positive increase. (↓) Positive decrease. (—) No significant change.

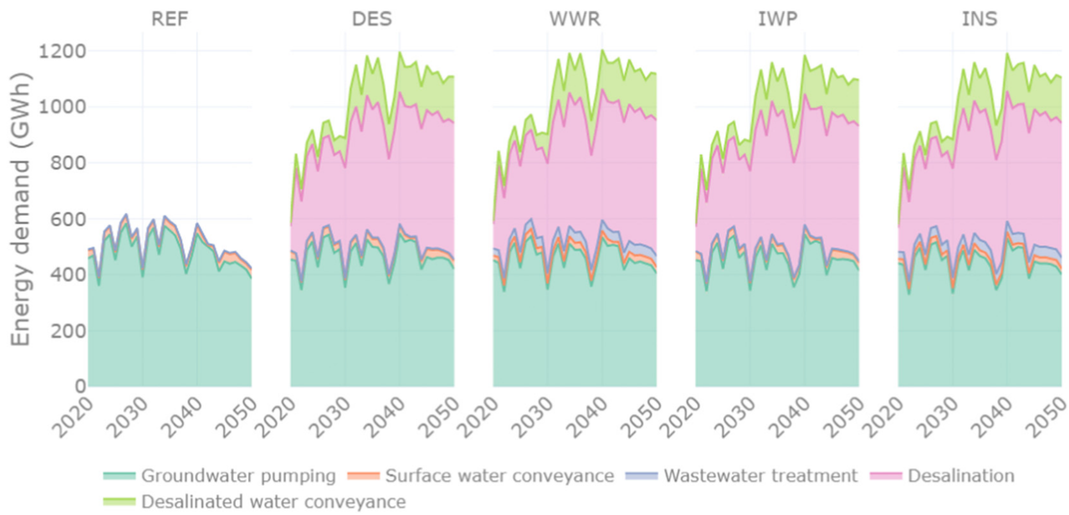

Fig. 12. Energy requirements for different activities across all scenarios: Reference (REF), Desalination(DES), Desalination with Wastewater Reuse (WWR), Increased Water Productivity (IWP) and Integrated Strategies (INS).

nexus approach. This novel framework initiated the dialogue with and between stakeholders on the robustness of the proposed solutions and modelling outcomes, which helped the gradual development of new perspectives, moving from single measures to an integrated strategy.

Decarbonisation strategies for the agricultural sector

Groundwater pumping is the second-highest energy user and butane fuels at least 20 % of the pumps in the region. As mentioned earlier, this part was developed in response to the request from the stakeholders to explore different decarbonization strategies and to shift from butane to solar pumping. This, we argue, is one of the benefits of implementing a participatory process at the sub-national level, where the level of complexity and conflicting interest among stakeholders is relatively low (if compared to the typical UNECE trans-boundaries nexus studies that involves diverse stakeholders from more than one country). The system cost and the environmental impacts considered in the analysis are from the government's perspective, not the end-users (farmers). The model takes into account the cost of the electricity generation from the national grid, butane subsidies, PV installation costs and the re-installation cost of PV for the units that will reach their lifetime before the end of the modelling period.

Looking at the green bars in Fig. 13, which represent the butane subsidy levels, it is noticeable that with the no phase-out the cumulative subsidy budget would reach a level of 6.6 billion MAD. The late phase-out by 2040 would reduce this cost to about 2.2 billion MAD while the early phase-out would further reduce the subsidies burden to about 1.2 billion MAD. In other words, the savings that the Moroccan government would gain are in the range of 4.4–5.4 billion MAD for early and late phase-out of butane respectively.

Moving away from butane would mean changing to the electricity from the grid or solar PV. The Moroccan grid is highly dependent on fossil fuels (International Energy Agency. IEA, 2019). The total emissions (in MtCO2) under each scenario were estimated based on the emission intensity of the grid and the butane pumps (see Table 3).

The best scenario for the government of Morocco is the one with the least cost and least emissions. Fig. 14 shows a comparison between the 12 sub-scenarios. The x-axis shows the total emission level (in MtCO2) under each scenario and the y-axis shows the total cost of the system (in million MAD) for the entire modelling period 2020–2050.

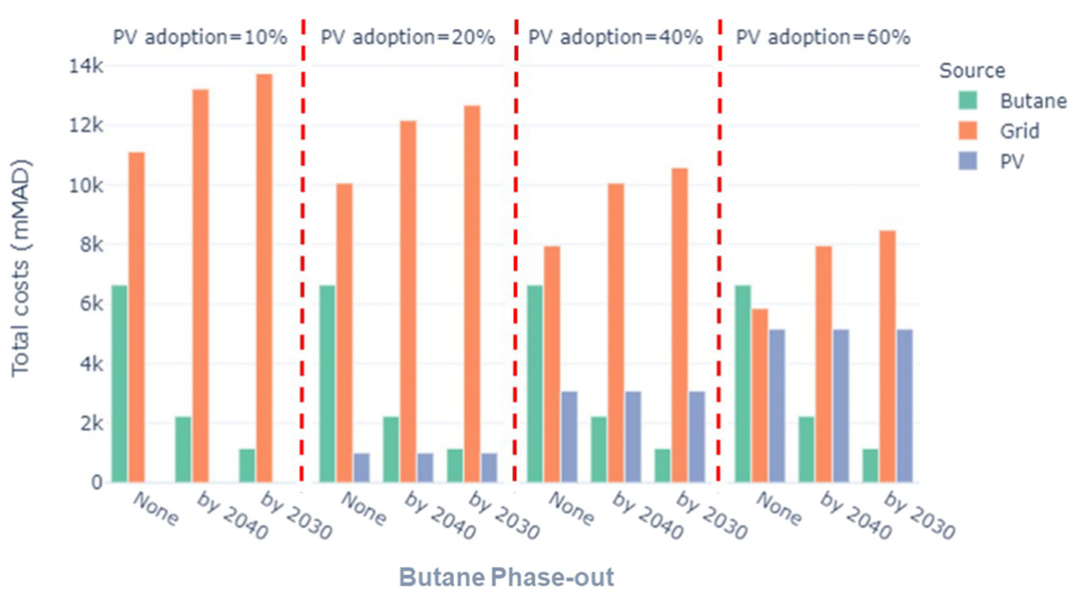

Fig. 13. Total system cost under different PV adaptation levels (10 %, 20 %, 40 % and 60 %) and different butane phase-out scenarios (None, by 2040 and by 2030). The bars shows the cost of each of the three pumping technologies (Butane, Grid and PV) in each case.

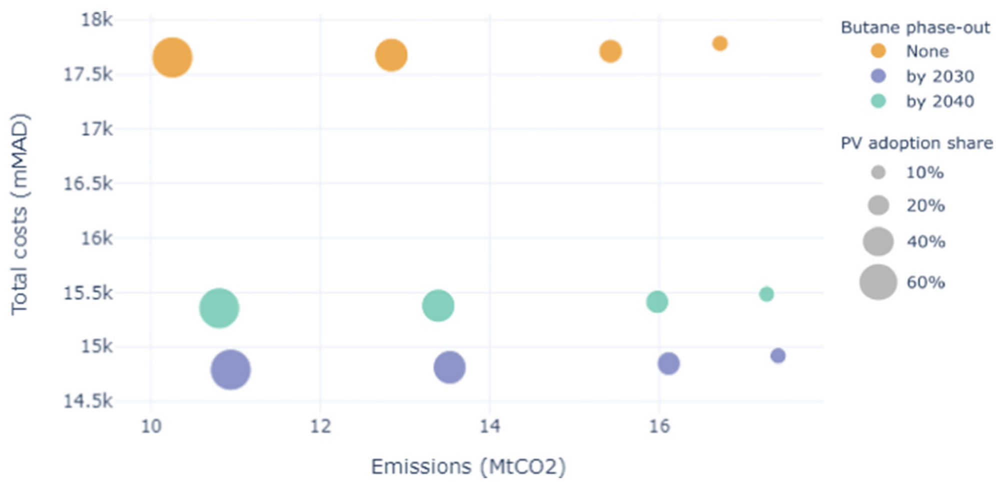

Fig. 14. Comparison between different butane phase-out and PV adaptation scenarios. The x-axis shows the total emissions and the y-axis shows the total system cost (butane, grid and PV).

The current situation (no phase-out scenario) will result in a very high system cost (about 17.8 bMAD) while the early phase-out by 2030 will have the lowest system cost (about 14.9 bMAD). The late phase out by 2040 will have a slightly higher cost (about 15.5 bMAD) if compared to the early phase-out. This is reflected in the distribution of the three colours in Fig. 14.

In terms of emissions, the higher the level of PV adoption, the lower the emissions level. This can be noticed by comparing the different bubble sizes and the corresponding emission levels. Again, the current situation (no phase-out scenario and 10 % PV level) would result in high environmental damages; we can say that this is the worst-case scenario. On the contrary, the best scenario would be to plan an early phase-out of butane by 2030 and invest in solar pumping to cover up to 60 % of the irrigated area. This will bring the Moroccan government savings of about 3 billion MAD and will cut about 6 MtCO2 emissions from the agricultural sector in Souss-Massa.

Conclusions

The SMB has great importance in the Moroccan agriculture sector. Several challenges are affecting the basin and its fertile land. The Souss-Massa basin's water supplies are insufficient to meet future demands due to overexploitation of groundwater resources and decline in precipitation. Qualitative and quantitative nexus analyses are needed to address intertwined WEF challenges in the region. Through a participatory process, the stakeholders mapped and prioritized four main challenges: a) **Water scarcity** and droughts. b) **Shift to energy independence**, c) **Agricultural productivity** and d) **Water quality**.

The impact of nexus challenges goes beyond the sector of origin. Therefore, engaging with stakeholders from different sectors is critical in nexus studies. The integration of the UNECE participatory nexus methodology with the Robust Decision Support process can be considered as one of the contributions of this study. Also, the implementation of this novel approach for the first time at the sub-national level allowed for a more in-depth engagement with local stakeholders and responding to their needs more effectively (e.g. exploring decarbonisation strategies in response to the energy sector's request). The introduction of a voting technique helped build consensus around the priority challenges and solutions. The vulnerability assessment helped to develop a common understanding of the key uncertainties and their impact on the system. In the work, climate change, energy import prices and changes in international markets for agricultural products came at the top of the list.

The nexus model of Souss-Massa was developed by soft-linking a GIS-based energy model with the water system model, WEAP. The nexus model allowed to quantitatively explore various potential solutions and to study four key scenarios that were jointly formulated with the stakeholders. The scenarios compare the Reference case, where no intervention is assumed, with three scenarios that examine different potential solutions (desalination, wastewater reuse and increased water productivity). Then, all measures were combined in an integrated strategies scenario. The sixth scenario explored 12 alternatives to decarbonise the agricultural sector in Souss-Massa by shifting from butane to solar PV for groundwater pumping.

From an energy point of view, desalination and groundwater pumping were shown to be the highest energy-intensive activities in Souss-Massa (primary agricultural phase). The seawater desalination plant would demand ~365–625 GWh annually to treat and convey water in the first and second phases of the project. The groundwater pumping in the entire basin would require ~400GWh annually. The Moroccan national grid supplies the biggest share of this demand. It will fully power the desalination plant and power ~70 % of the groundwater pumps. Knowing that the Moroccan grid is heavily based on fossil fuels, the environmental and climate impacts of these activities should be mitigated. This underlines the importance of renewable energy to power the desalination plant as well as to power groundwater pumps. The phase-out of butane used for groundwater pumping and shift to solar PV has the potential to be an economically and environmentally competitive strategy. However, this shift towards PV pumping systems should go hand in hand with effective measures (e.g. monitoring systems) to avoid groundwater overexploitation with an energy source that has no unit costs.

This study shows that no one solution can address all nexus challenges. Instead, a combination of solutions should be integrated to increase synergies and minimize trade-offs. Enhancing water supply and utilising new sources (e.g. desalination and wastewater reuse), improving agricultural practices and reducing the overuse of water resources were explored as an integrated strategy scenario. This has been shown to bring various benefits such as reducing the depletion of the groundwater resources, especially in the Chtouka aquifer, increasing the productivity of the crops and reducing the unmet water demand for agriculture and domestic sectors.

As with any research, our study has limitations that we would like to highlight. The decision matrix developed for the first part does not account for the cost of different strategies or scenarios. This would be an important element to consider in any future work. The sensitivity of the analysis to different phase-out years and different PV shares were explored. However, as uncertainties exist in other parameters that could affect the outcomes of the analysis like the CAPEX, OPEX, the discount rate, future work may attempt to expand the presented sensitivity analysis. Data paucity is a limitation throughout most input data of the water model, particularly regarding tributaries flow. Also, the simplified climate change representation in this study should be enhanced in any future work and consider the impact of different climate change projections. Additionally, the lack of low and medium voltage transmission lines maps hindered the investigation of the spatial distribution of PV investments, which also hindered exploring the economics of selling the excess electricity to the national grid during off-peak hours.

The findings of this study were used to inform the decision-making process and fed into several local workshops on the WEF nexus to understand and assess the trade-off of the proposed strategies. Furthermore, the outcomes of this study were used to showcase the WEF nexus application in the Near East and North Africa (NENA) region through a series of webinars hosted by FAO (Food and Agriculture Organization of the United Nations. FAO, 2021) and led by the authors. This bridges the gap between science and policy dimensions and shares the lessons learned from the SMB with the larger nexus community. This study can pave the way for future research agenda such as: using mixed methods to inform groundwater governance in water scarce regions; assessing the impact of climate change on water resources availability in the NENA region; or understanding the role of WEF nexus in achieving SDG 2 on zero hunger, SDG6 on water and sanitation and SDG7 on sustainable energy.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Nexus mapping exercise

This study used the participatory process explained in Qualitative analysis. Qualitative analysis to interact with the stakeholders. This process is workshop oriented and most of the inputs from the stakeholders are stimulated through group work on specific tasks. The final results are then incorporated into the following steps of the process and the study. In the first nexus dialogue workshop which is focused on identifying the nexus challenges, the stakeholder inputs were collected through two main activities. In the first activity the stakeholders worked in groups representing their sectors (e.g. water, energy, food, environment). Each group was tasked to answer the following questions using the sectoral map shown in Fig. A1:

1. What are the key future plans in your sector? Select 3–4 main objectives.
2. To achieve these objectives, what resources required from other sectors? Draw a line showing the flow of the resources from one sector to the other.
3. How does these goals affect other sectors? Draw a line showing the impact from one sector on the other.
4. In a scale of 1 (low) to 5 (high) prioritize the importance of main interactions identified in steps 2 and 3.

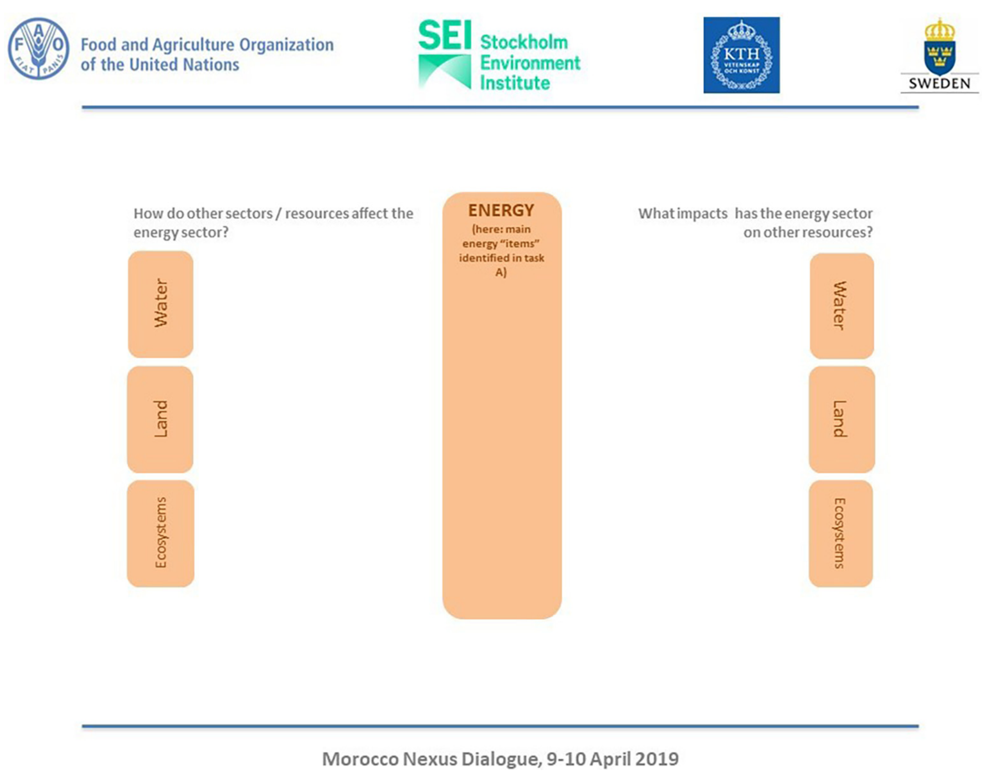

Fig. A1. Sample of the sectoral nexus mapping.

In the second activity the stakeholders were divided in mixed groups from different sectors. Each group was asked to answer the following questions:
In the light of the discussion in the previous session:

1. Present the interactions with the high importance (scored 4 and 5 in step 4 of the previous session) to the rest of the group, and add them on the graph.
2. Are there any additional interactions that are missing? Please add them.
3. After adding all the interactions, what are the most pressing interactions? And why? Rank them in a scale from 1 to 5.
4. We will call those high important interactions as (nexus challenges). Prepare a separate graph for the top 3 interactions.

Each group has to then report back and present the issues that they have identified to other groups. After a detailed panel discussion on each the challenges, the common issues qualified to the next step where participants used the voting technique (mentemeter) to prioritise the nexus challenges as shown in Appendix C.

Appendix B

Energy model: mathematical correlations/representation:

- **Electricity requirement for groundwater pumping and surface water conveyance:**

Energy for water pumping can be expressed as the energy required to lift water from groundwater sources and to overcome friction in pipes, pumps, and other elements of the distribution system used for conveyance across the land surface. Electrical energy, or electricity (kWh), is expended when a unit volume (m^3) of water passes through a pump during its operation (Ahlfeld & Laverty, 2011).

The electricity demand depends on the efficiency of the pump, the pipeline length and diameter, pipe material roughness or friction factor, and the volumetric demand for water. As shown in the following function for electricity demand, E_D (kWh):

$$E_D = f(d, Q, P, t, f_l)$$

where d is the distance through which the water is to be lifted, Q is the required volumetric amount of water for pumping, P is the pressure required at the point of use, t is the time over which the water is pumped (assuming a constant head), and f_l is the friction loss along with the distance within the distribution system.

The calculation of the electricity demand (E_D in kWh) for water pumping can be calculated as follows:

$$E_D(\text{KWh}) = \frac{\text{SSWD}(m^3) * \rho\left(\frac{kg}{m^3}\right) * g\left(\frac{m}{s^2}\right) * \text{TDH}_{gw}(m)}{\text{PP}_{eff}(\%)} * \frac{1}{3600\left(\frac{s}{h}\right)} * \frac{1}{1000\left(\frac{W}{kW}\right)}$$

where Seasonal Scheme Water Demand $SSWD$ (m^3) is defined as the total volume of water required over a selected season, $\rho\left(\frac{kg}{m^3}\right)$ is the density of water, $g\left(\frac{m}{s^2}\right)$ is the acceleration of gravity (9.81 m/s2) and $\frac{1}{3600\left(\frac{s}{h}\right)} * \frac{1}{1000\left(\frac{W}{kW}\right)}$ are factors to convert from Joule units to kWh.

Moreover, $TDH_{gw}(m)$ represents the Total Dynamic Head and $PP_{eff}(\%)$ accounts for the Pumping Plant efficiency. The calculation of the Total Dynamic Head is estimated using the following equation:

$$\text{TDH}_{gw}\ (m) = \text{EL}(m) + \text{SL}(m) + \text{OP}(m) + \text{FL}(m)$$

where $EL(m)$ is the Elevation Lift, $SL(m)$ expresses the Suction Lift, $OP(m)$ stands for Operating Pressure and accounts for the pressure needed based on the application and conveyance system, and $FL(m)$ expresses the Friction Losses in the piping systems.
The overall power required for pumping water is determined as:

$$\text{PD}_{gw}\ \text{kW} = 9.81 * \text{discharge}\left(\frac{m^3}{s}\right) * \text{TDH}_{gw}(m) * \text{PP}_{eff}(\%)$$

where $discharge\left(\frac{m^3}{s}\right)$ is the Peak Scheme Water Demand expressed in $\left(\frac{m^3}{s}\right)$.

- **Electricity requirement for seawater desalination:**

The energy requirements for desalination is estimated

$$E_{desal}\ (\text{KWh}) = Q_{desal}(m^3) * e_{desal}\left(\frac{\text{KWh}}{m^3}\right)$$

where Q_{desal} is the total volume of the desalinated water (obtained from WEAP) in (m^3) and e_{desal} is the energy intensity per volume of desalinated water in $\left(\frac{KWh}{m^3}\right)$.

- **Electricity requirement for wastewater treatment:**

Similarly, the energy requirements for wastewater treatment

$$E_{desal}\ (\text{KWh}) = Q_{wwt}(m^3) * e_{wwt}\left(\frac{\text{KWh}}{m^3}\right)$$

where Q_{wwt} is the total volume of the treated water (obtained from WEAP) in (m^3) and e_{wwt} is the energy intensity per volume of desalinated water in $\left(\frac{KWh}{m^3}\right)$.

• Decarbonization scenarios: butane, grid and PV calculations

The calculations in this part of the analysis are based on the groundwater demand (estimated using WEAP) and the energy for groundwater pumping that was estimated in the previous step. It is worth mentioning that WEAP aggregates all the demand into nodes, but in this step, we distribute the electricity demand into the cropland layer to have a better spatial distribution. This is done proportionally by the area fraction of the irrigated area of each polygon of the cropland layer (see the GIS section).

Once we have the spatial distribution of the pumping demand, and since we do not have detailed information of the locations of each technology (Butane, PV and grid), we assumed that three technologies are used in each location but with different shares: PV:10 %, Butane: 20 % and Grid: 70 %. This is the starting point for the reference case and we will explore how the share of these technologies will change under different phase-out scenarios as explained below (see Scenarios).

The first step is to estimate the electricity demand for each technology in each location, which is based on:

$$\mathrm{E_{tech}\ (KWh)} = \frac{\mathrm{Peak\ Load\ (KW)^{*}Share_{tech}\ (\%)}}{\mathrm{pump\ efficiency_{tech}\ (\%)}}$$

where E_{tech} is the required electricity for pumping from each technology in (KWh), *Peak Load* (KW) is the peak pumping load in each month for each location in (KW), $Share_{tech}$ is the share of each technology at each location in(%) (note that the share will change in each year according to the phase-out scenarios) and $pump\ efficiency_{tech}$ represents the pump efficiency for each technology.
After estimating the electricity required for each technology, we estimated the amount of butane required to supply this electricity.

$$\mathrm{Fuel.Requirment_{butane}\ (Kg)} = \mathrm{E_{butane}\ (KWh)^{*}Specific.Energy_{butane}}\left(\frac{\mathrm{kg}}{\mathrm{KWh}}\right)$$

where E_{butane} is the electricity required to run butane pumps in (KWh) and the $Specific.\ Energy_{butane}$ gives how much electricity can be produced by burning 1 Kg of butane, which is about (12.7 kWh).
This will help us estimate the cost of butane, knowing that in this analysis we are interested in the government perspective in terms of the subsidy cost. There we will base the calculations on the subsidies that are currently about (80 MAD) per bottle of Butane, which has 12 kg. Note that this is estimated for each location and for each year for the entire modelling period from 2020 to 2050.

$$\mathrm{Cost_{butane}\ (MAD)} = \mathrm{Fuel.Requirment_{butane}\ (Kg)^{*}Subsidy_{butane}}\left(\frac{\mathrm{MAD}}{\mathrm{Kg}}\right)$$

The cost of electricity from the grid also represents the cost of generation (from the government's perspective). This was based on the electricity generation mix in Morocco in 2017 and the technology cost. Therefore, the grid cost at each location was estimated as follows:

$$\mathrm{Cost_{grid}\ (MAD)} = \mathrm{E_{grid}\ (KWh)^{*}gen.cost_{grid}}\left(\frac{\mathrm{MAD}}{\mathrm{KWh}}\right)$$

where E_{grid} is the required electricity for pumping from each technology in (KWh) and $gen.\ cost_{grid}$ is the generation cost of one unit of electricity in $\left(\frac{MAD}{KWh}\right)$.

The emissions released from each technology can be estimated using:

$$\mathrm{emissions_{tech}\ (Mt.CO_2)} = \mathrm{E_{tech}\ (KWh)^{*}ef_{tech}}\left(\frac{\mathrm{Kg.CO_2}}{\mathrm{KWh}}\right)$$

where E_{tech} is the required electricity for pumping from each technology in (KWh) and ef_{tech} is the emission factor of each technology in $\left(\frac{Kg.CO_2}{KWh}\right)$. A conversion factor is used to report all emissions in ($Mt.\ CO_2$).

And the total emissions are simply the sum of the grid emissions and butane pumps emissions in each year:

$$\mathrm{Total\ emissions\ (Mt.CO_2)} = \mathrm{emissions_{grid}\ (Mt.CO_2)^{*}emissions_{butane}\ (Mt.CO_2)}$$

Now moving to solar PV, we need first to estimate the Capacity Factor of PV in each location. To do this, we will use the monthly solar irradiation map that gives us the radiation in $\left(\frac{KJ}{m2.day}\right)$ and we estimate the monthly CF as follows:

$$CF_{PV} = \frac{solar.irradiation_{PV}\left(\frac{KJ}{m2.day}\right)}{24\left(\frac{h}{day}\right)^{*}60\left(\frac{min}{h}\right)^{*}60\left(\frac{sec}{\mathrm{min}}\right)}$$

Then the required solar PV capacity in each month will be:

$$\mathrm{Capacity_{PV}\ (MW)} = \frac{\mathrm{Peak\ Load\ (KW)^{*}Share_{PV}\ (\%)}}{\mathrm{CF_{PV}{}^{*}1000}\left(\frac{\mathrm{KW}}{\mathrm{MW}}\right)}$$

After knowing the required installed capacity in each location in each month, we will take the maximum monthly capacity to estimate the investment cost of solar PV. Bear in mind that the solar panels have a specific lifetime, which is assumed in this analysis as 25 years, this means we need to take into consideration the investment and the reinvestment costs. This will depend on the year of initial investment, which can vary from one location to another.

$$\mathrm{Cost_{PV}\ (MAD)} = \left[\mathrm{Capacity_{PV}\ (KW)}^{*}\mathrm{CapitalCost_{PV}}\left(\frac{\mathrm{MAD}}{\mathrm{KW}}\right)\right] + \left[\mathrm{Reinvest.Capacity_{PV}\ (KW)}^{*}\mathrm{CapitalCost_{PV}}\left(\frac{\mathrm{MAD}}{\mathrm{KW}}\right)\right]$$

where $Cost_{PV}$ is the capital cost (CAPEX) for PV in MAD or million MAD, $Capacity_{PV}$ is the required capacity of PV to be installed in each location in (KW) or (MW), $Reinvest.\ Capacity_{PV}$ is the reinvestment required to replace the solar panels in some locations after their lifetime period and finally the $CapitalCost_{PV}$ in $\left(\frac{MAD}{KW}\right)$. It is worth mentioning that in this analysis, a learning curve of PV was assumed which shows an annual decline in technology cost to reach a 40 % drop by 2040 (compare to the cost in 2020). The operating and maintenance cost of PV was assumed as 1 % of the CAPEX.

Finally, the total cost of the system is:

$$\mathrm{Total\ Costs\ (MAD)} = \mathrm{Cost_{butane}\ (MAD)} + \mathrm{Cost_{grid}\ (MAD)} + \mathrm{Cost_{PV}\ (MAD)}$$

Additional explanations of the modelling steps can be found in the script's documentation.

Water model: calibration

The first step in calibrating the WEAP model was to select a historical period of record that includes concurrent input and observation data that cover a period long enough to capture the range of conditions (wet and dry) within a basin. The main input data for the WEAP model include climate data, the observation data are gaged streamflow.

Streamflow

Streamflow data for several locations within the Souss-Massa basin were obtained from the Plan Directeur d'Amenagement Integre des Resource en Eau (PDAIRE) for the 30-year historical period from 1977 to 2006 as summarized in Table B1.

Table B1
Percent of period with monthly streamflow observations.

Location	Beginning date	End date	Total number of records	Percent missing
Wadi Amaghouz	Apr 1976	Aug 2006	353	2 %
Wadi Ibergnaten	Nov 1976	Aug 2006	346	4 %
Wadi Ifni	Nov 1979	Aug 2006	309	14 %
Wadi Immikki	Mar 1980	Aug 2006	305	15 %
Wadi Issen at Aguenza	Jan 1980	Aug 2006	308	14 %
Wadi Lamdad	Sept 1976	Aug 2006	249	30 %
Wadi Souss at Ait Melloul	Sept 1989	Aug 2006	48	87 %

These data include stations that are mostly on tributaries to the Souss and Massa rivers as shown in Fig. B1. Thus, represent locations that are minimally influenced by water management (i.e. diversions) within the basin. These observations were used to calibrate the natural hydrology of the basin.

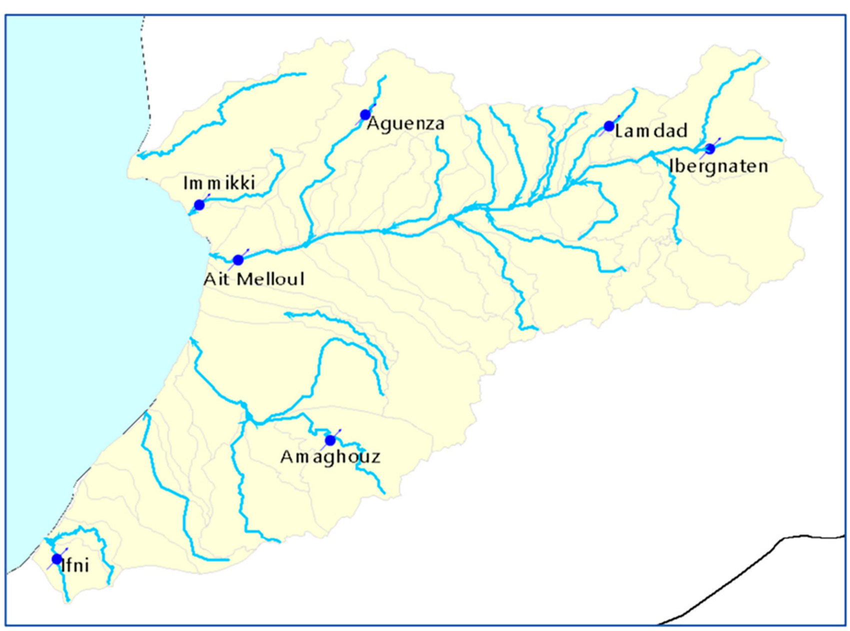

Fig. B1. Streamflow stations used in calibrating WEAP hydrology simulation.

Climate

Monthly precipitation data was collected from the meteorological stations within the SMB for a historical period dating back to 1966 and extending to 2017. Table B2 below gives a summary of these data and shows the concentration of records that overlap with the observed (1977–2006) streamflow data obtained from the PDAIRE.

Table B2
Meteorological stations within the Souss-Massa River Basin.

Station name	Beginning date	End date	Total number of records (1977–2006)	Percent missing (1977–2006)
Aguenza	Sept 1981	Aug 2017	300	17 %
Amaghouz	Sept 1976	Aug 2017	360	0 %
Amsoul	Sept 1978	Aug 2017	336	7 %
Iguidi	Sept 1988	Aug 2017	146	59 %
Immikki	Sept 1983	Aug 2017	276	23 %
Immerguen	Sept 1970	Aug 2017	360	0 %
Lamdad	Sept 1995	Aug 2017	132	63 %
Ouijjane	Sept 1976	Aug 2017	360	0 %
Pont Aoulouz	Sept 1966	Aug 2017	324	10 %
Pont Taroudant	Sept 1966	Aug 2017	360	0 %
Taliouin	Sept 1988	Aug 2017	109	70 %
Tamri	Sept 1991	Aug 2017	180	50 %
Tassila	Sept 2000	Aug 2017	72	80 %

The lack of adequate resources to operate and maintain the metrological stations in the SMB has resulted in inconsistent and occasionally sparse data collection. For this reason, the thirty-five-year period between 1977 and 2006 was chosen as the period for model calibration and validation, because it contains the highest concentration of climate data over the period of record as shown in Table B2 and Fig. B2.

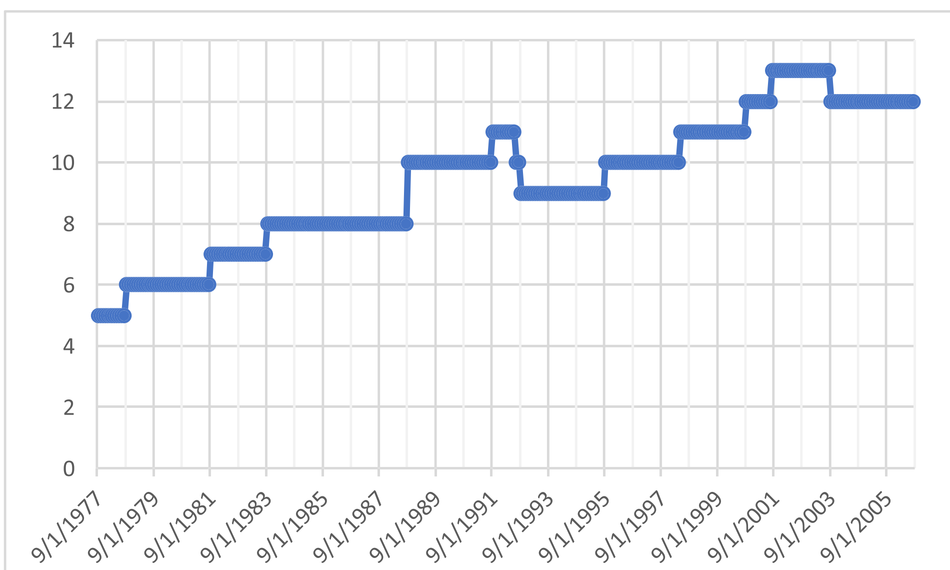

Fig. B2. Number of active meteorological stations, 1976–2010.

The Souss-Massa WEAP model was calibrated to historical streamflows using a combination of manual methods and the optimization tool PEST, which is an advanced software tool used for model calibration, parameter estimation and predictive uncertainty analysis. Calibration parameters included the Nash-Suttcliffe Efficiency (NSE) factor, which measures how well simulated monthly flows match observed streamflow data; the percent bias (PBIAS), which provides a percentage measure for the average difference between simulated and observed data; and the ratio of the simulated and observed standard deviation (SDR), indicates how well the simulated data capture the observed range of streamflow values. In general, simulated streamflow is judged to be satisfactory when NSE > 0.5, PBIAS <25 %, and 0.8 < SDR < 1.2. Calibration results are summarized in Table B3.

Table B3
Calibration metrics.

	NSE	PBIAS	SDR
Wadi Amaghouz	0.36	−3 %	0.76
Wadi Ibergnaten	0.22	3 %	1.28
Wadi Ifni	0.08	−3 %	0.80
Wadi Tamraght at Immikki	0.55	1 %	1.05
Wadi Issen at Aguenza	0.03	−6 %	1.23

Table B3 (*continued*)

	NSE	PBIAS	SDR
Wadi Lamdad	0.53	1 %	1.48
Wadi Souss at Ait Melloul	0.64	−2 %	0.96

While some of the objective function values presented in Table B3 are relatively poor, the overall results are generally acceptable. Where there are observed data available that can be considered reasonably representative of flow conditions, the WEAP model is able to approximate the frequency characteristics. The sometimes poor values for NSE and PBIAS are likely to be partly associated with inadequate rainfall data to accurately quantify individual monthly rainfall totals at the catchment scale, the effects of different spatial distributions of rainfall for the same monthly spatially-averaged depth and the effects of different temporal distributions of rainfall for the same monthly depth (i.e. different daily distributions). None of these types of uncertainty can be resolved with a monthly time-step model applied in an area the size of the Souss-Massa basin where many parts of the basin are inadequately gauged.

These issues of data uncertainty can be seen by looking at the percentage of precipitation that arrives at one example streamflow gage, Wadi Issen at Aguenza (Fig. B3). The simulated values fall within the range of expected percentages and the month-to-month changes in these percentages are generally gradual, which is to be expected with a hydrological model that accounts for runoff attenuation and soil water storage. On the other hand, the observed values are much more erratic, do not exhibit a recognizable seasonal pattern, and occasionally reach unreasonable values. This suggests that there are inconsistencies between the precipitation and streamflow data. It should be the long-term goal to resolve these issues by conducting a thorough examination of the input and observation data. However, it is our contention that the calibration is sufficient for the purposes of using the model to conduct scenario analysis as a means of exploring potential management strategies for the basin.

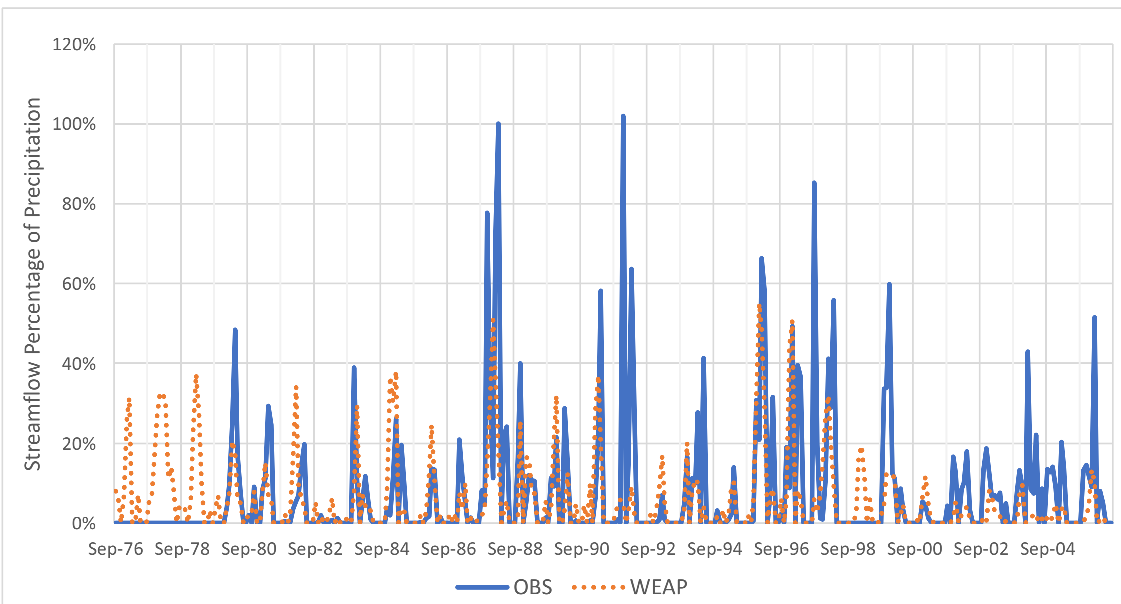

Fig. B3. Simulated and observed monthly streamflow at Aguenza as a percentage of precipitation.

Water model: key inputs

Table B4
Irrigated cropped areas within the Souss-Massa basin.

	Equipped area (ha)	Irrigated area (ha)	Cereals	Fodder	Trees	Vegetables	
						Open field	Greenhouse
Tamri/Tamraght province							
PMH Tamri_Tamraght	1560	1560	1030	296	234	0	
Souss province							
El Guerdane	10,000	10,000	4900	900	2700	500	1000
G1 Aoulouz	4500	4500	3600	225	675	0	
Modern Privé de Souss	40,400	20,200	5252	7878	3838	3232	
Modern Public du Souss Amont	6120	6120	1591	2387	1163	979	
Modern Public Issen	8560	8560	1370	856	5050	1284	
PMH Anti Atlas Ait Baha	2120	2120	1399	0	572	148	
PMH Anti Atlas Ighrem	2540	2540	1803	0	635	102	
PMH de Montagne Souss	5655	5655	170	2149	3110	226	
PMH Taliouine Tifnoute	7940	7940	7384	318	0	238	

(continued on next page)

Table B4 (*continued*)

	Equipped area (ha)	Irrigated area (ha)	Cereals	Fodder	Trees	Vegetables	
						Open field	Greenhouse
Traditionnel nn Réahbilité 2 Tranche	9220	9220	277	3504	4979	461	
Traditionnel Rehabilite du Souss Amont	1350	1350	41	513	743	54	
Traditionnel Réhabilité du Souss Perimetres Diff	1540	1540	46	601	832	62	
Traditionnel Rehabilité Issen	4440	3019	453	362	2113	91	
Traditionnel Rehabilité Souss Amont 1 Tranche	10,160	10,160	305	3861	5588	406	
Massa province							
Modern Privé Massa	7500	7500	1050	1350	525	2059	2516
Modern Public Massa	18,050	9928	1390	1787	695	3452	2604
Moderne Privé Chtouka	7500	7500	4950	0	2025	236	289
Oughzifen	130	111	19	19	11	62	
Tassila	1200	660	112	112	66	370	
Tiznit province							
PMH de Tiznit	5300	5300	1044	848	3095	313	
Total	155,785	125,482	38,185	27,965	38,649	15,274	5409

Appendix C

Results of the qualitative analysis

The following figures show additional results of the stakeholders mapping of the nexus challenges.

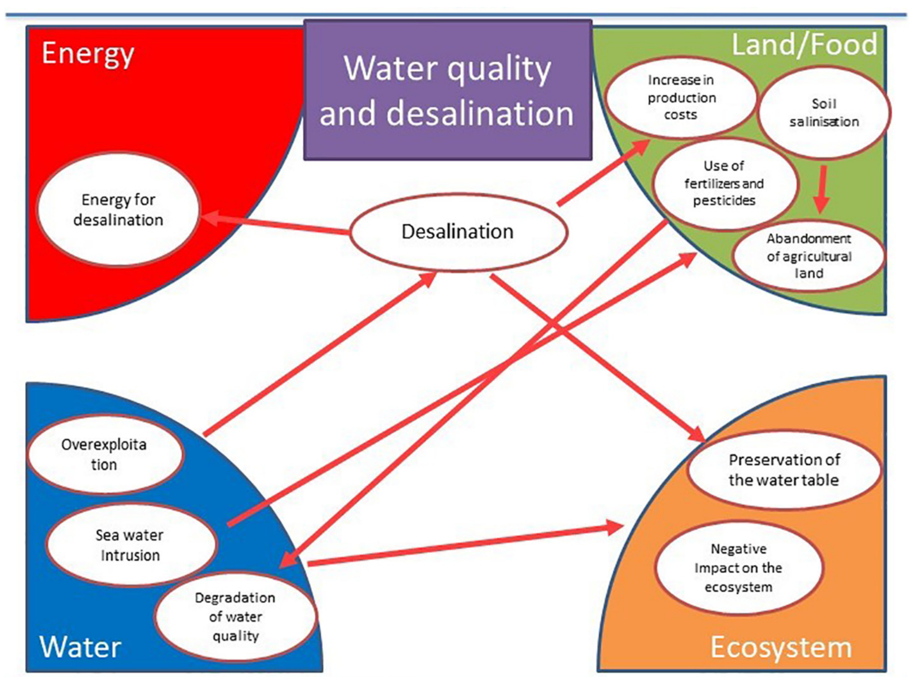

Fig. C1. Mapping of the nexus challenge - water quality.

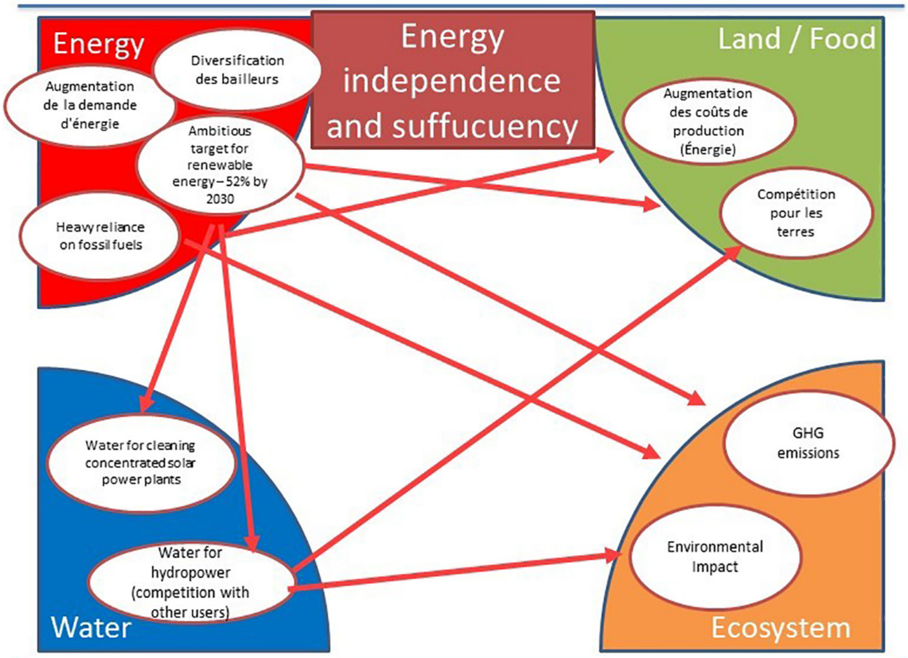

Fig. C2. Mapping of the nexus challenge - energy sufficiency.

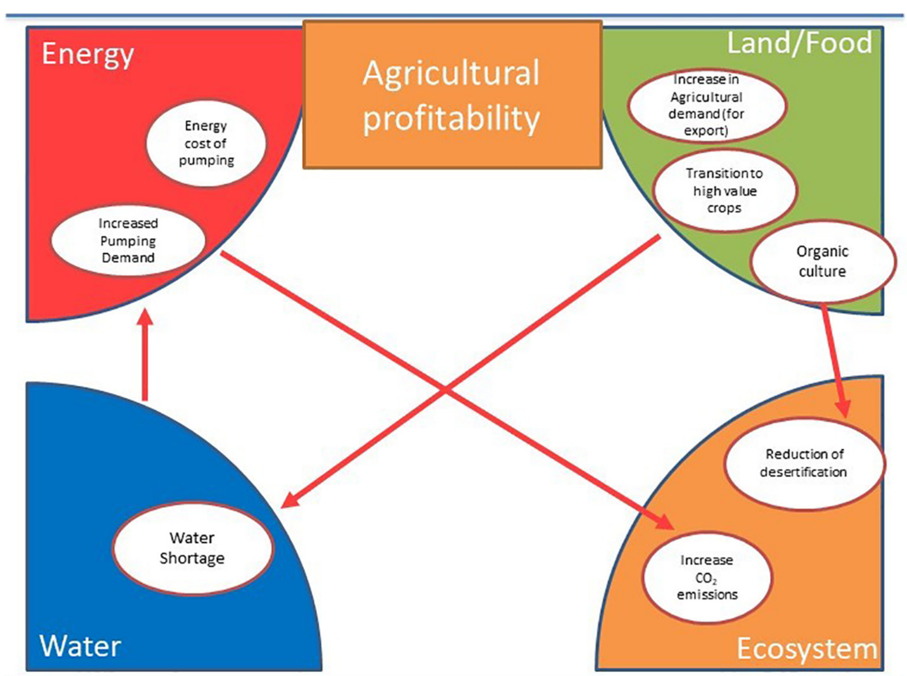

Fig. C3. Mapping of the nexus challenge - agriculture productivity.

Results of the quantitative analysis

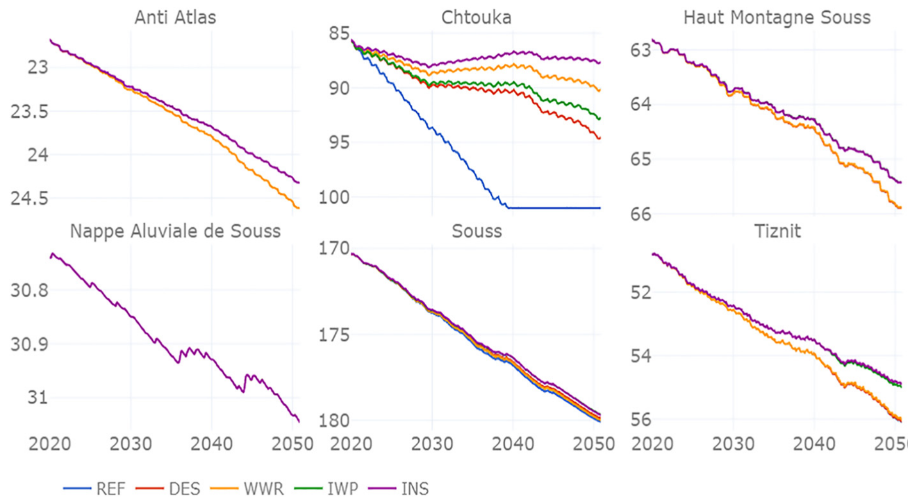

Fig. C4. Change in groundwater table depth (m) for all aquifers across all scenarios: Reference (REF), Desalination(DES), Desalination with Wastewater Reuse (WWR), Increased Water Productivity (IWP) and Integrated Strategies (INS).

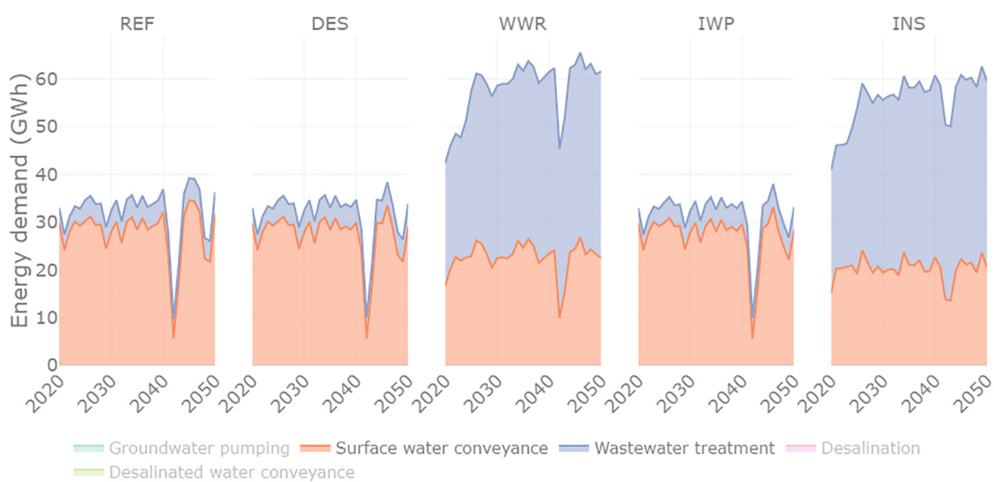

Fig. C5. Energy requirements for surface water conveyance, wastewater treatment and wastewater reuse conveyance across all scenarios: Reference (REF), Desalination(DES), Desalination with Wastewater Reuse (WWR), Increased Water Productivity (IWP) and Integrated Strategies (INS).

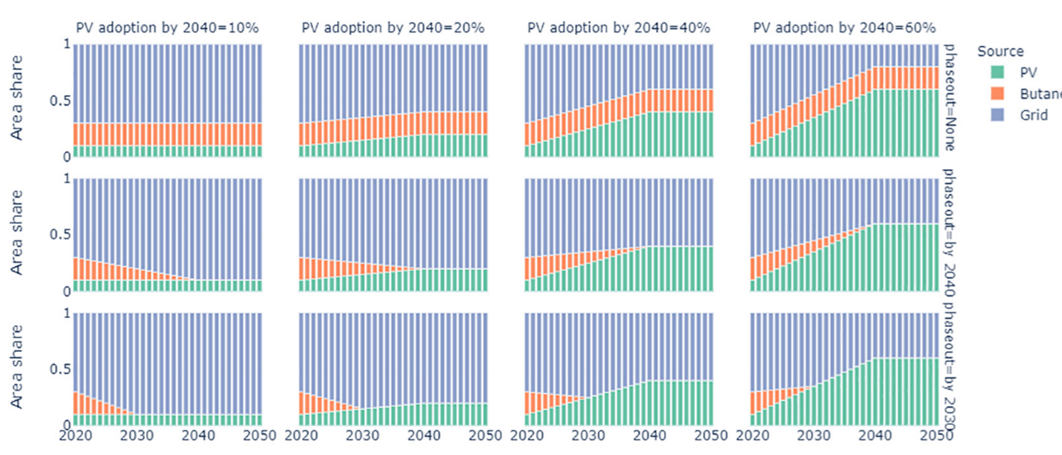

Fig. C6. Dynamics explored in each sub-scenario based on the change in butane phase-out year and PV adoption level. The rows show different phase-out years of 2050 at the top, then 2040 and 2030 in the lower row. While the columns show the three PV levels of 10 % to the far left and 60 % to the far right.

Table C1
Results of the decarbonization scenarios.

Butane phaseout	Pv adoption	PV Capex (mMAD)	Grid cost (mMAD)	Butane subsidy (mMAD)	Total costs (mMAD)	Total emissions ($MtCO_2$)
None	10 %	25	11,115	6643	17,784	16.72
None	20 %	1007	10,063	6643	17,713	15.43
None	40 %	3077	7957	6643	17,677	12.84
None	60 %	5160	5852	6643	17,655	10.26
By 2040	10 %	25	13,220	2240	15,485	17.27
By 2040	20 %	1007	12,168	2240	15,414	15.98
By 2040	40 %	3077	10,063	2240	15,379	13.39
By 2040	60 %	5160	7957	2240	15,357	10.81
By 2030	10 %	25	13,739	1155	14,919	17.41
By 2030	20 %	1007	12,686	1155	14,848	16.11
By 2030	40 %	3077	10,581	1155	14,813	13.53
By 2030	60 %	5160	8476	1155	14,791	10.94

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